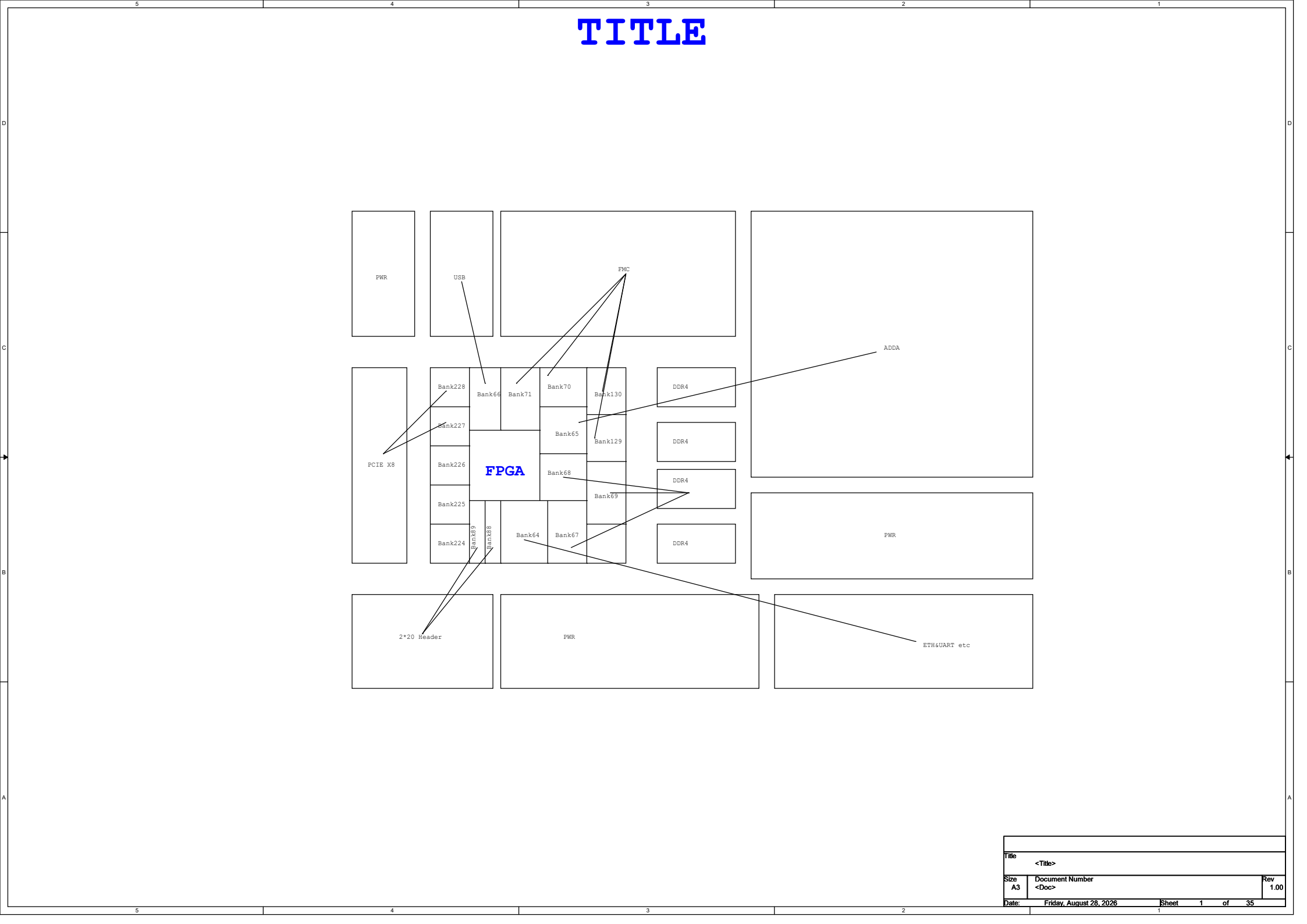


TITLE

The diagram illustrates a hardware system architecture centered around an **FPGA**. The FPGA is connected to various components and memory banks:

- Top Section:** PWR, USB, FMC, and a large block labeled ADDA.
- Left Section:** PCIE X8, 2*20 Reader, and PWR.
- Right Section:** ETH&UART etc, PWR, and three DDR4 modules.
- Central Section:** The FPGA is connected to several memory banks: Bank224, Bank225, Bank226, Bank227, Bank228, Bank64, Bank65, Bank66, Bank67, Bank68, Bank69, Bank70, Bank71, Bank129, and Bank130.

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TITLE

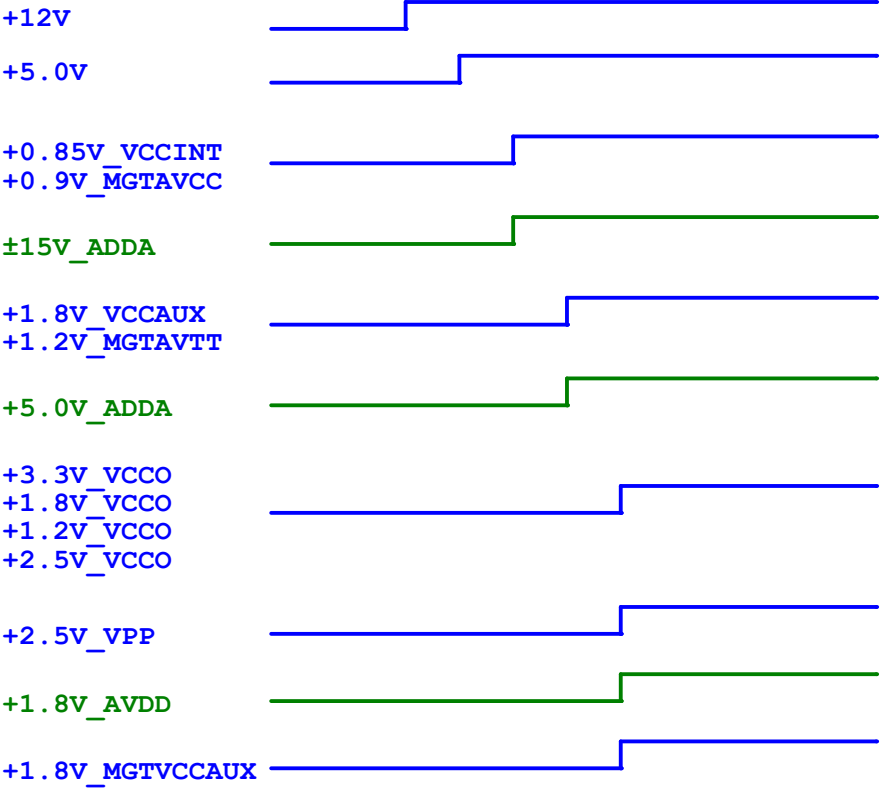
The diagram illustrates a hardware system architecture centered around an **FPGA**. The FPGA is connected to various components and memory banks:

- Power and I/O:** PWR, USB, FMC, PCIE X8, 2*20 Reader, PWR, ETH&UART etc.
- Memory:** Bank224, Bank225, Bank226, Bank227, Bank228, Bank64, Bank65, Bank66, Bank67, Bank68, Bank69, Bank70, Bank71, Bank129, Bank130.
- Other:** ADDA, DDR4 (three modules).

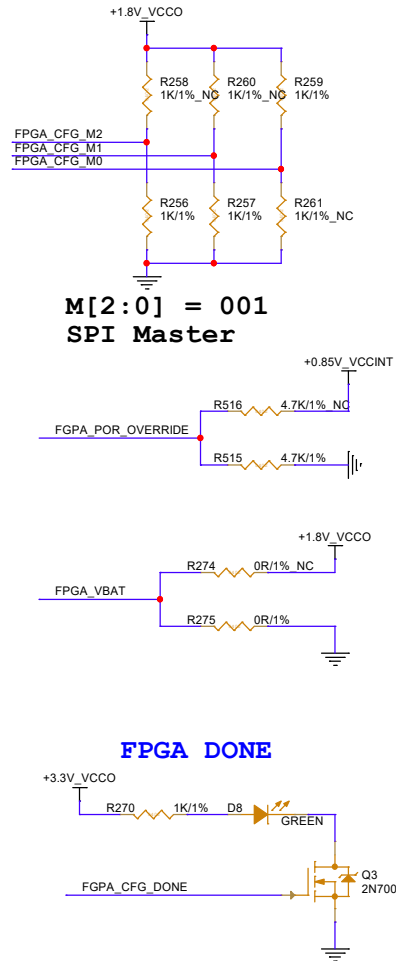
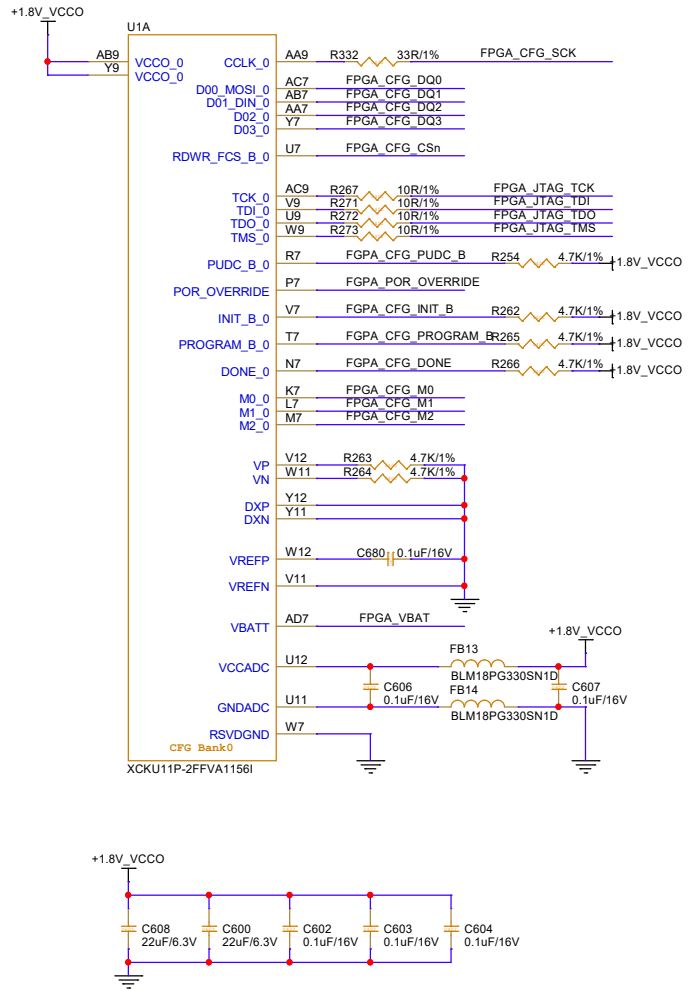
Legend:

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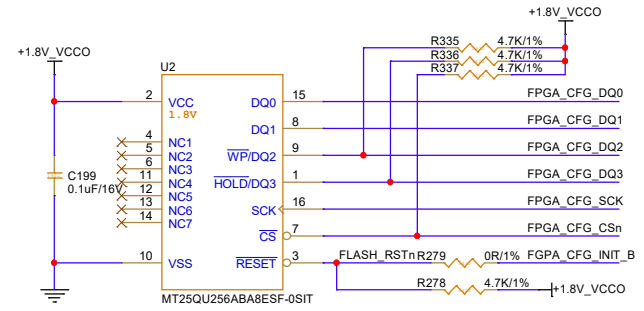
Diagram



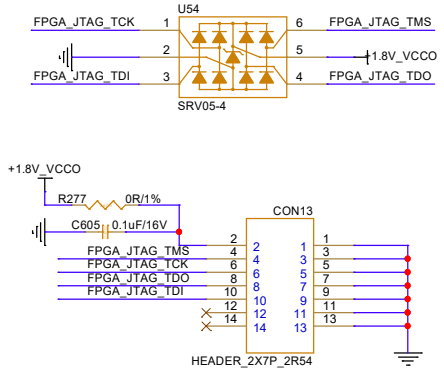
FPGA_CFG



M[2:0] = 001
SPI Master



NOR Flash 256Mb

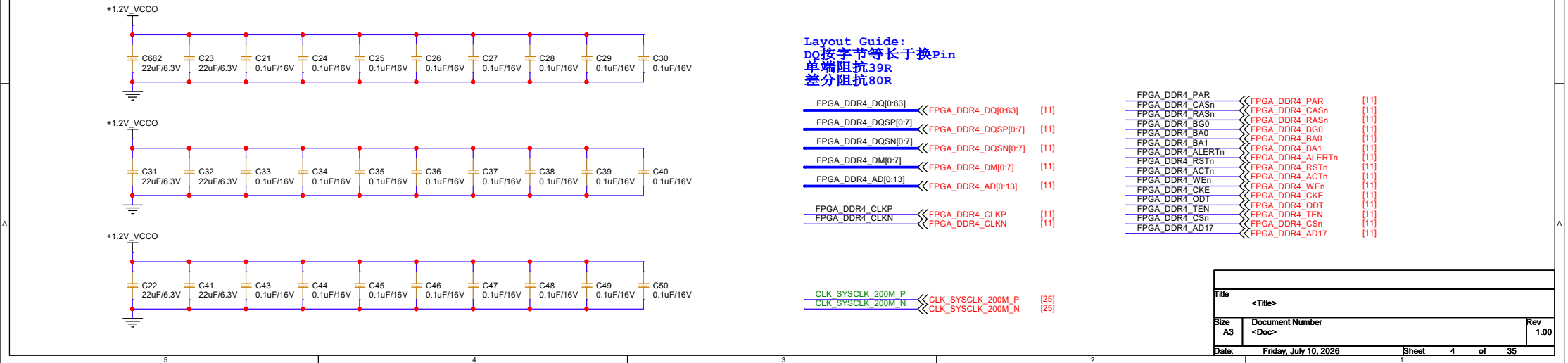
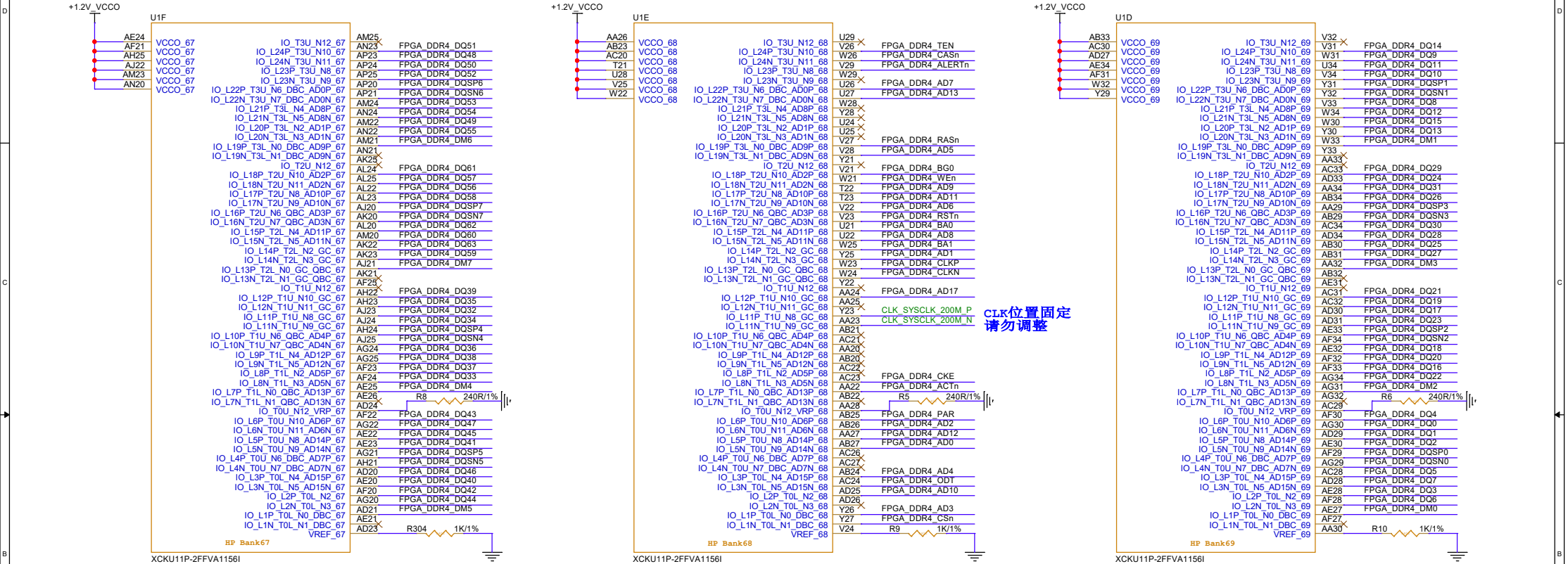


JTAG Port

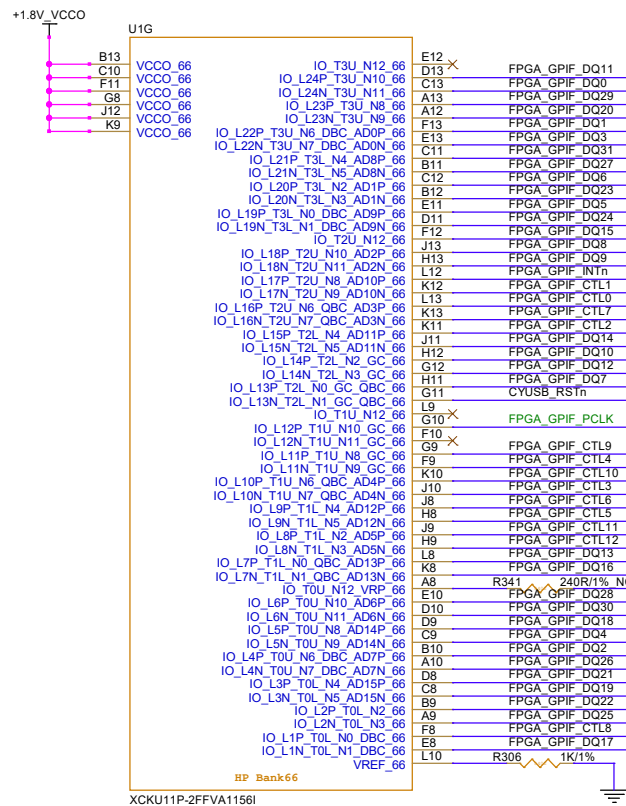
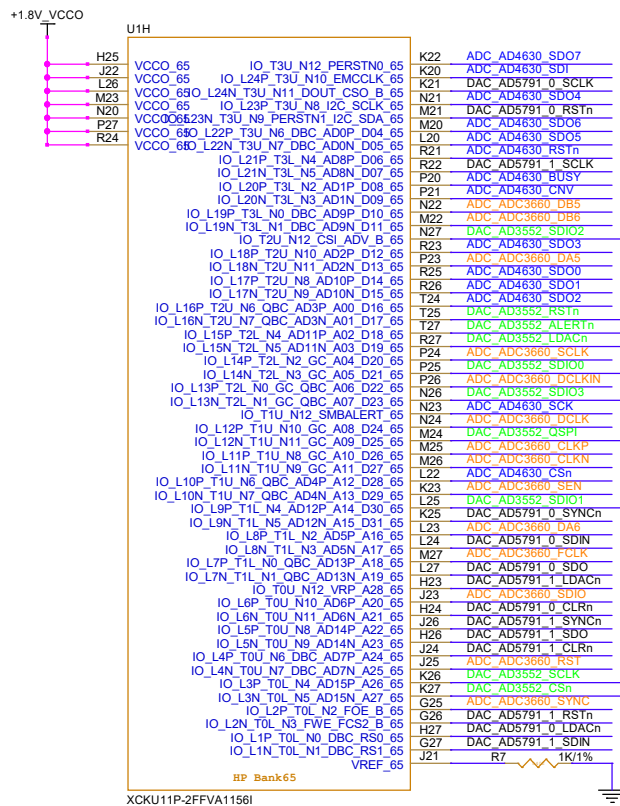
修订日志:
1、修改网络错误,原1.8V VCC0改成+1.8V VCC0 - 2026-09-03
2、CON13信号左右对调,符合JTAG标准定义 - 2026-09-03

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FPGA_DDRMC



FPGA_ADDA&USB



PCLK位置固定
请勿调整

1.8V IO Level

Low Speed DAC
AD5792_CH0

DAC_AD5791_0_SYNCn	DAC_AD5791_0_SYNCn	[15]
DAC_AD5791_0_SCLK	DAC_AD5791_0_SCLK	[15]
DAC_AD5791_0_SDO	DAC_AD5791_0_SDO	[15]
DAC_AD5791_0_SDIN	DAC_AD5791_0_SDIN	[15]
DAC_AD5791_0_LDACn	DAC_AD5791_0_LDACn	[15]
DAC_AD5791_0_CLRn	DAC_AD5791_0_CLRn	[15]
DAC_AD5791_0_RSTn	DAC_AD5791_0_RSTn	[15]

Low Speed DAC
AD5792 CH1

DAC_AD5791_1_SYNCn	DAC_AD5791_1_SYNCn	[15]
DAC_AD5791_1_SCLK	DAC_AD5791_1_SCLK	[15]
DAC_AD5791_1_SDO	DAC_AD5791_1_SDO	[15]
DAC_AD5791_1_SDIN	DAC_AD5791_1_SDIN	[15]
DAC_AD5791_1_LDACn	DAC_AD5791_1_LDACn	[15]
DAC_AD5791_1_CLRn	DAC_AD5791_1_CLRn	[15]
DAC_AD5791_1_RSTn	DAC_AD5791_1_RSTn	[15]

High Speed DAC
AD5792

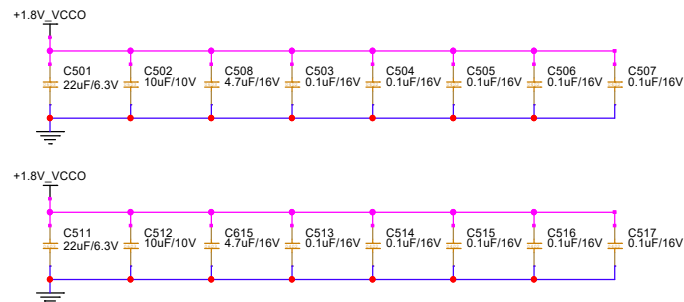
DAC_AD3552_SDI0[0:3]	≡	DAC_AD3552_SDI0[0:3]	[16]
DAC_AD3552_SCLK	≡	DAC_AD3552_SCLK	[16]
DAC_AD3552_LDACn	≡	DAC_AD3552_LDACn	[16]
DAC_AD3552_RSTn	≡	DAC_AD3552_RSTn	[16]
DAC_AD3552_ALERTn	≡	DAC_AD3552_ALERTn	[16]
DAC_AD3552_QSPi	≡	DAC_AD3552_QSPi	[16]
DAC_AD3552_CSn	≡	DAC_AD3552_CSn	[16]

Low Speed ADC AD4630-24

ADC_AD4630_SDO[0:7]	≡≡ ADC_AD4630_SDO[0:7]	[21]
ADC_AD4630_CS _n	≡≡ ADC_AD4630_CS _n	[21]
ADC_AD4630_SDI	≡≡ ADC_AD4630_SDI	[21]
ADC_AD4630_SCK	≡≡ ADC_AD4630_SCK	[21]
ADC_AD4630_CNV	≡≡ ADC_AD4630_CNV	[21]
ADC_AD4630_BUSY	≡≡ ADC_AD4630_BUSY	[21]
ADC_AD4630_RST _n	≡≡ ADC_AD4630_RST _n	[21]

High Speed ADC ADC3660

ADC_ADC3660_DCLKIN	ADC_ADC3660_DCLKIN	[24]
ADC_ADC3660_DA6	ADC_ADC3660_DA6	[24]
ADC_ADC3660_DA5	ADC_ADC3660_DA5	[24]
ADC_ADC3660_DB6	ADC_ADC3660_DB6	[24]
ADC_ADC3660_DB5	ADC_ADC3660_DB5	[24]
ADC_ADC3660_DCLK	ADC_ADC3660_DCLK	[24]
ADC_ADC3660_FCLK	ADC_ADC3660_FCLK	[24]
ADC_ADC3660_SCLK	ADC_ADC3660_SCLK	[24]
ADC_ADC3660_SCLK	ADC_ADC3660_SCLK	[24]
ADC_ADC3660_RST	ADC_ADC3660_RST	[24]
ADC_ADC3660_CLKP	ADC_ADC3660_CLKP	[24]
ADC_ADC3660_CLKN	ADC_ADC3660_CLKN	[24]



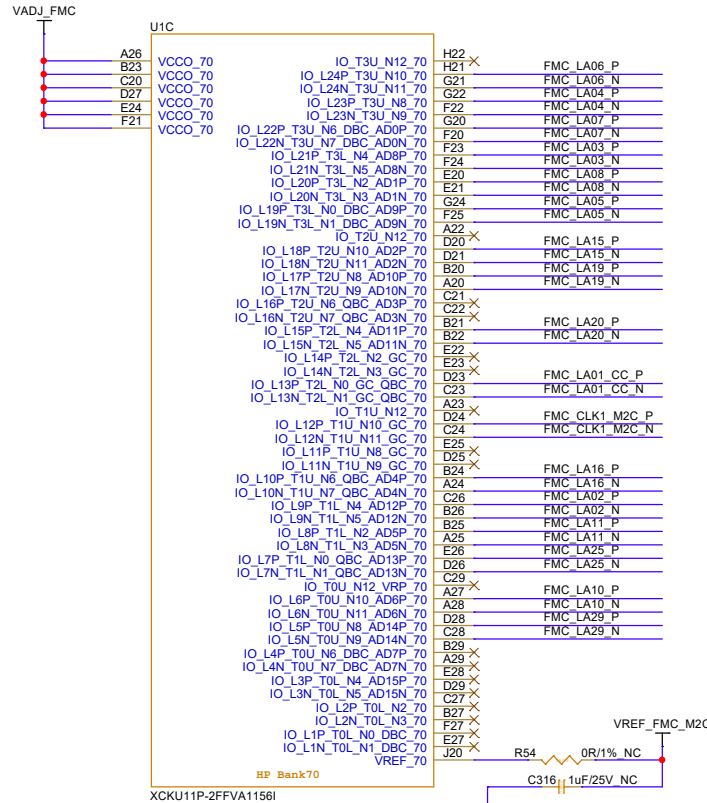
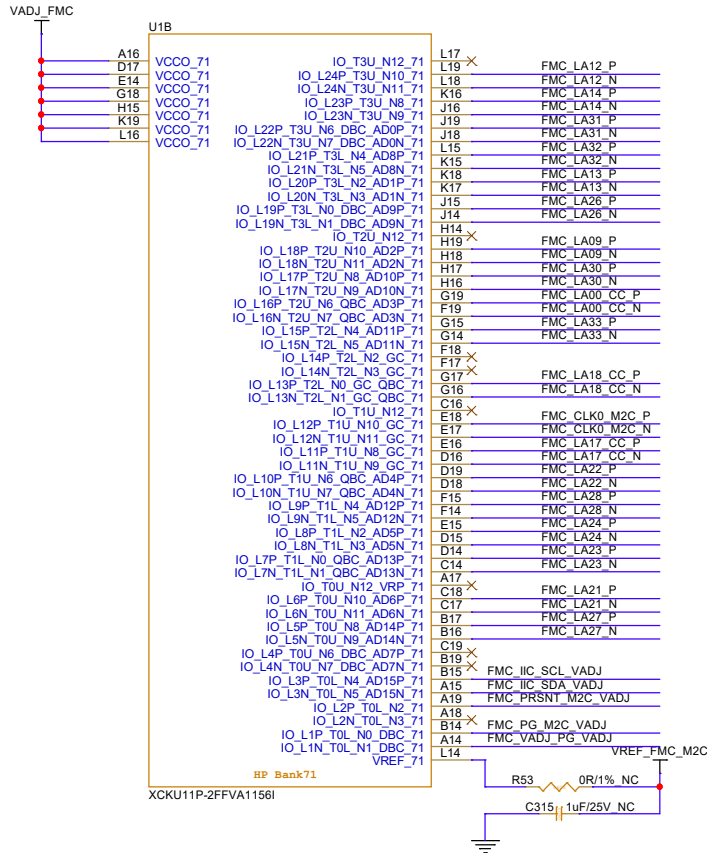
DQ&PCLK组内等长5mil

CTL&INT组内等长5mil

1.8V Signal
单端阻抗50R @100Mbps

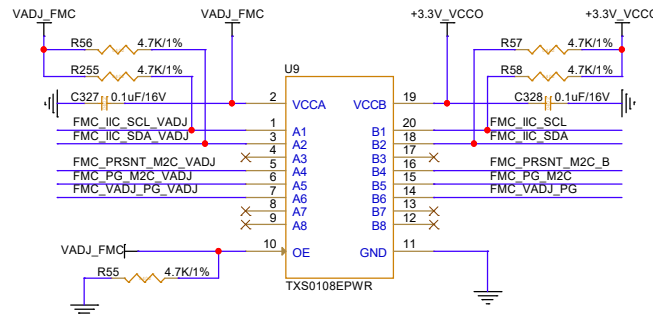
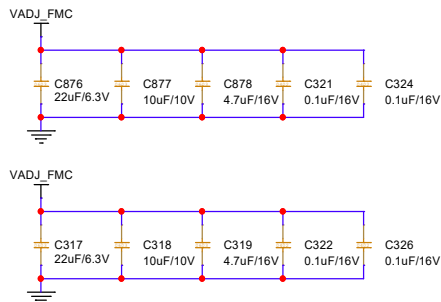
FPGA_GPIF_DQ[0:31]	◀◀FPGA_GPIF_DQ[0:31]	[12]
FPGA_GPIF_PCLK	◀◀FPGA_GPIF_PCLK	[12]
FPGA_GPIF_CTL[0:12]	◀◀FPGA_GPIF_CTL[0:12]	[12]
FPGA_GPIF_INTn	◀◀FPGA_GPIF_INTn	[12]
CYUSB_RSTn	◀◀CYUSB_RSTn	[12]

FPGA_HP10_FMC



VADJ_FMC
电源域

FMC LA00 CC_P	FMC LA00 CC_P	[33]
FMC LA00 CC_N	FMC LA00 CC_N	[33]
FMC LA01 CC_P	FMC LA01 CC_P	[33]
FMC LA01 CC_N	FMC LA01 CC_N	[33]
FMC LA02 P	FMC LA02 P	[33]
FMC LA02 N	FMC LA02 N	[33]
FMC LA03 P	FMC LA03 P	[33]
FMC LA03 N	FMC LA03 N	[33]
FMC LA04 P	FMC LA04 P	[33]
FMC LA04 N	FMC LA04 N	[33]
FMC LA05 P	FMC LA05 P	[33]
FMC LA05 N	FMC LA05 N	[33]
FMC LA06 P	FMC LA06 P	[33]
FMC LA06 N	FMC LA06 N	[33]
FMC LA07 P	FMC LA07 P	[33]
FMC LA07 N	FMC LA07 N	[33]
FMC LA08 P	FMC LA08 P	[33]
FMC LA08 N	FMC LA08 N	[33]
FMC LA09 P	FMC LA09 P	[33]
FMC LA09 N	FMC LA09 N	[33]
FMC LA10 P	FMC LA10 P	[33]
FMC LA10 N	FMC LA10 N	[33]
FMC LA11 P	FMC LA11 P	[33]
FMC LA11 N	FMC LA11 N	[33]
FMC LA12 P	FMC LA12 P	[33]
FMC LA12 N	FMC LA12 N	[33]
FMC LA13 P	FMC LA13 P	[33]
FMC LA13 N	FMC LA13 N	[33]
FMC LA14 P	FMC LA14 P	[33]
FMC LA14 N	FMC LA14 N	[33]
FMC LA15 P	FMC LA15 P	[33]
FMC LA15 N	FMC LA15 N	[33]
FMC LA16 P	FMC LA16 P	[33]
FMC LA16 N	FMC LA16 N	[33]
FMC LA17 CC_P	FMC LA17 CC_P	[33]
FMC LA17 CC_N	FMC LA17 CC_N	[33]
FMC LA18 CC_P	FMC LA18 CC_P	[33]
FMC LA18 CC_N	FMC LA18 CC_N	[33]
FMC LA19 P	FMC LA19 P	[33]
FMC LA19 N	FMC LA19 N	[33]
FMC LA20 P	FMC LA20 P	[33]
FMC LA20 N	FMC LA20 N	[33]
FMC LA21 P	FMC LA21 P	[33]
FMC LA21 N	FMC LA21 N	[33]
FMC LA22 P	FMC LA22 P	[33]
FMC LA22 N	FMC LA22 N	[33]
FMC LA23 P	FMC LA23 P	[33]
FMC LA23 N	FMC LA23 N	[33]
FMC LA24 P	FMC LA24 P	[33]
FMC LA24 N	FMC LA24 N	[33]
FMC LA25 P	FMC LA25 P	[33]
FMC LA25 N	FMC LA25 N	[33]
FMC LA26 P	FMC LA26 P	[33]
FMC LA26 N	FMC LA26 N	[33]
FMC LA27 P	FMC LA27 P	[33]
FMC LA27 N	FMC LA27 N	[33]
FMC LA28 P	FMC LA28 P	[33]
FMC LA28 N	FMC LA28 N	[33]
FMC LA29 P	FMC LA29 P	[33]
FMC LA29 N	FMC LA29 N	[33]
FMC LA30 P	FMC LA30 P	[33]
FMC LA30 N	FMC LA30 N	[33]
FMC LA31 P	FMC LA31 P	[33]
FMC LA31 N	FMC LA31 N	[33]
FMC LA32 P	FMC LA32 P	[33]
FMC LA32 N	FMC LA32 N	[33]
FMC LA33 P	FMC LA33 P	[33]
FMC LA33 N	FMC LA33 N	[33]
FMC CLK0 M2C_P	FMC CLK0 M2C_P	[33]
FMC CLK0 M2C_N	FMC CLK0 M2C_N	[33]
FMC CLK1 M2C_P	FMC CLK1 M2C_P	[33]
FMC CLK1 M2C_N	FMC CLK1 M2C_N	[33]

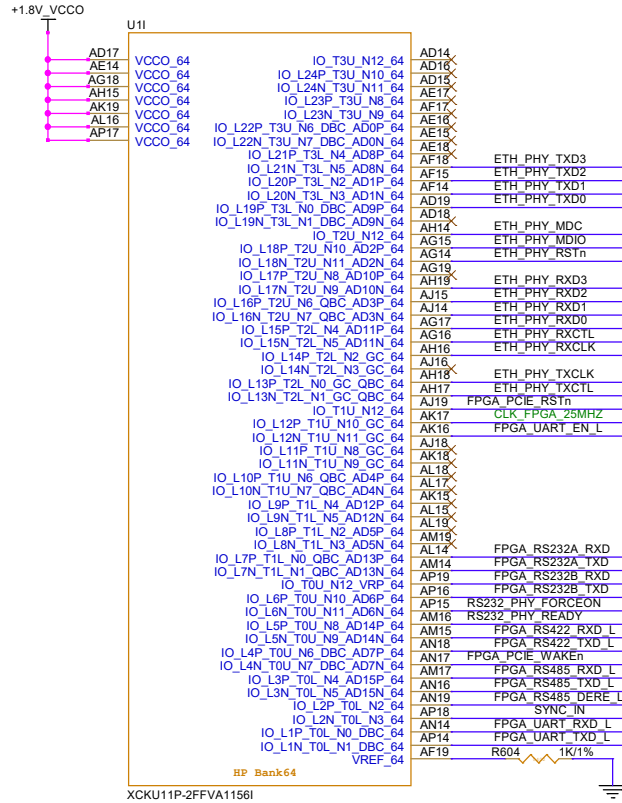
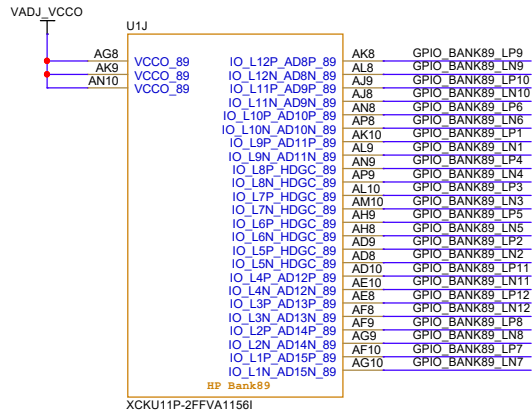
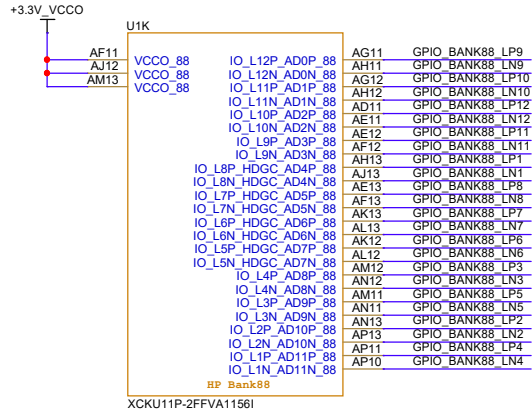


3.3V
电源域

FMC IIC_SCL	FMC IIC_SCL	[33]
FMC IIC_SDA	FMC IIC_SDA	[33]
FMC PRSNT_M2C_B	FMC PRSNT_M2C_B	[33]
FMC PG_M2C	FMC PG_M2C	[33]
FMC VADJ_PG	FMC VADJ_PG	[33]

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FPGA_OTHER



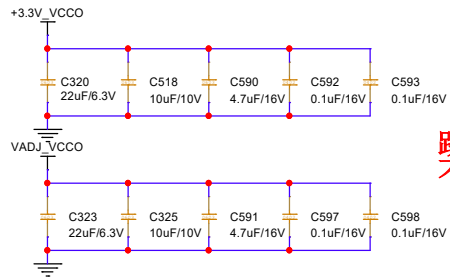
CLK位置固定
请勿调整

LSDAC_CH1_50ROUT_BYPASS << LSDAC_CH1_50ROUT_BYPASS

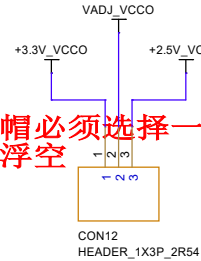
LSDAC_CH2_50ROUT_BYPASS << LSDAC_CH2_50ROUT_BYPASS

HSDAC_CH1_50ROUT_BYPASS << HSDAC_CH1_50ROUT_BYPASS

HSDAC_CH2_50ROUT_BYPASS << HSDAC_CH2_50ROUT_BYPASS



跳线帽必须选择一个电压
不可浮空



BANK88 3.3V Level

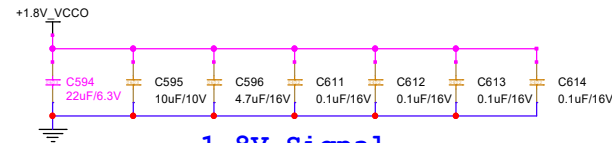
GPIO_BANK88_LP[1:12] << GPIO_BANK88_LP[1:12] [34]
GPIO_BANK88_LN[1:12] << GPIO_BANK88_LN[1:12] [34]

BANK89 2.5V or 3.3V Level

GPIO_BANK89_LP[1:12] << GPIO_BANK89_LP[1:12] [34]
GPIO_BANK89_LN[1:12] << GPIO_BANK89_LN[1:12] [34]

ETH 1.8V

ETH_PHY_TXD[0:3] << ETH_PHY_TXD[0:3] [13]
ETH_PHY_TXCTL << ETH_PHY_TXCTL [13]
ETH_PHY_TXCLK << ETH_PHY_TXCLK [13]
ETH_PHY_RXD[0:3] << ETH_PHY_RXD[0:3] [13]
ETH_PHY_RXCTL << ETH_PHY_RXCTL [13]
ETH_PHY_RXCLK << ETH_PHY_RXCLK [13]
ETH_PHY_MDC << ETH_PHY_MDC [13]
ETH_PHY_MDIO << ETH_PHY_MDIO [13]
ETH_PHY_RSTn << ETH_PHY_RSTn [13]



1.8V Signal

FPGA_PCIE_WAKEn << FPGA_PCIE_WAKEn [35]
FPGA_PCIE_RSTn << FPGA_PCIE_RSTn [35]
SYNC_IN << SYNC_IN [34]
FPGA_RS232A_RXD << FPGA_RS232A_RXD [14]
FPGA_RS232A_TXD << FPGA_RS232A_TXD [14]
FPGA_RS232B_RXD << FPGA_RS232B_RXD [14]
FPGA_RS232B_TXD << FPGA_RS232B_TXD [14]
RS232_PHY_FORCEON << RS232_PHY_FORCEON [14]
RS232_PHY_READY << RS232_PHY_READY [14]

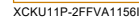
1.8V Level Signal

FPGA_RS485_DERE_L << FPGA_RS485_DERE_L [14]
FPGA_RS485_TXD_L << FPGA_RS485_TXD_L [14]
FPGA_RS485_RXD_L << FPGA_RS485_RXD_L [14]
FPGA_RS422_RXD_L << FPGA_RS422_RXD_L [14]
FPGA_RS422_TXD_L << FPGA_RS422_TXD_L [14]
FPGA_UART_RXD_L << FPGA_UART_RXD_L [14]
FPGA_UART_TXD_L << FPGA_UART_TXD_L [14]
FPGA_UART_EN_L << FPGA_UART_EN_L [14]

CLK_FPGA_25MHZ << CLK_FPGA_25MHZ [25]

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接FMC的GT Lane
AC电容均放置于FMC模块中



XCKU11P-2FFVA1156I



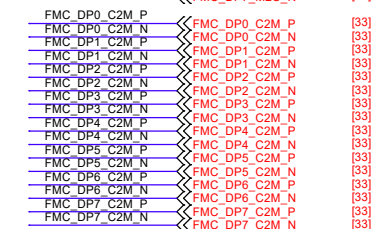
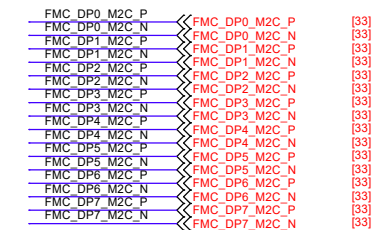
XCKU11P-2FFVA1156

U1M



FMC_GTCLK1_M2C_P
 FMC_GTCLK1_M2C_N
 FMC_GTCLK0_M2C_P
 FMC_GTCLK0_M2C_N

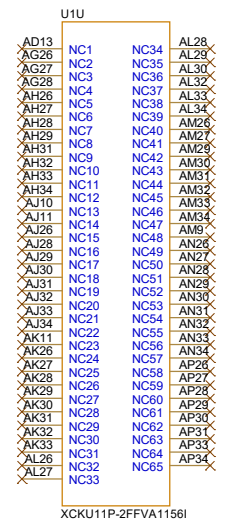
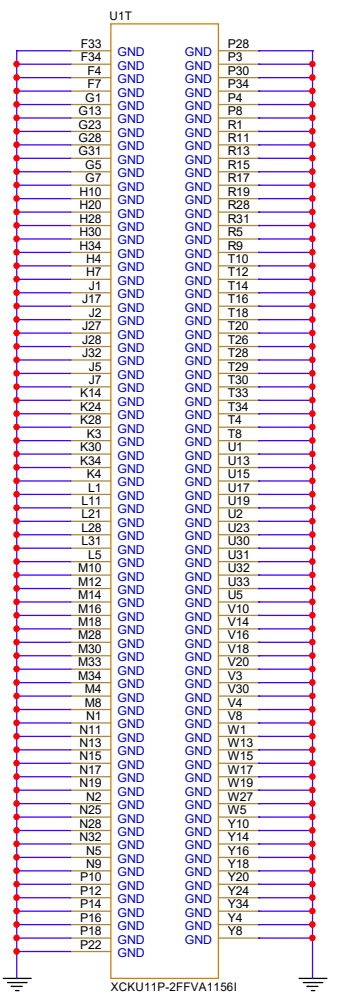
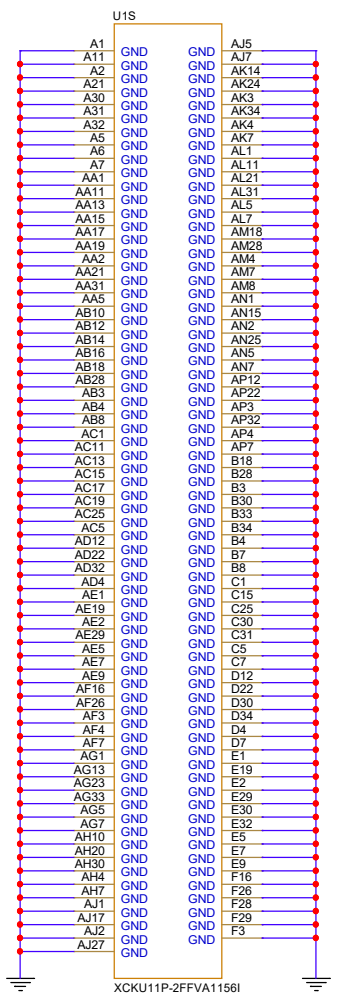
FMC_GTCLK1_M2C_P [33]
 FMC_GTCLK1_M2C_N [33]
 FMC_GTCLK0_M2C_P [33]
 FMC_GTCLK0_M2C_N [33]



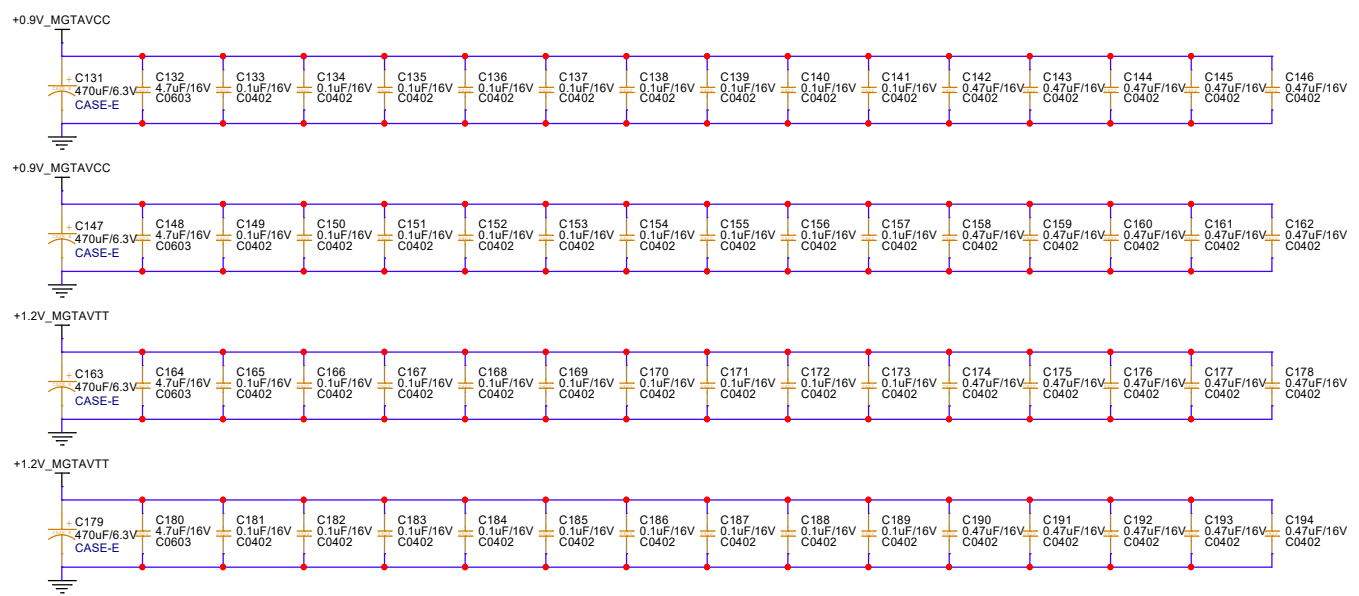
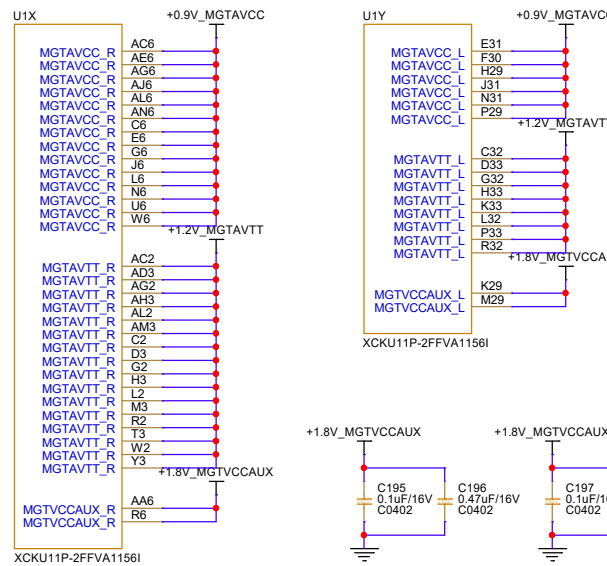
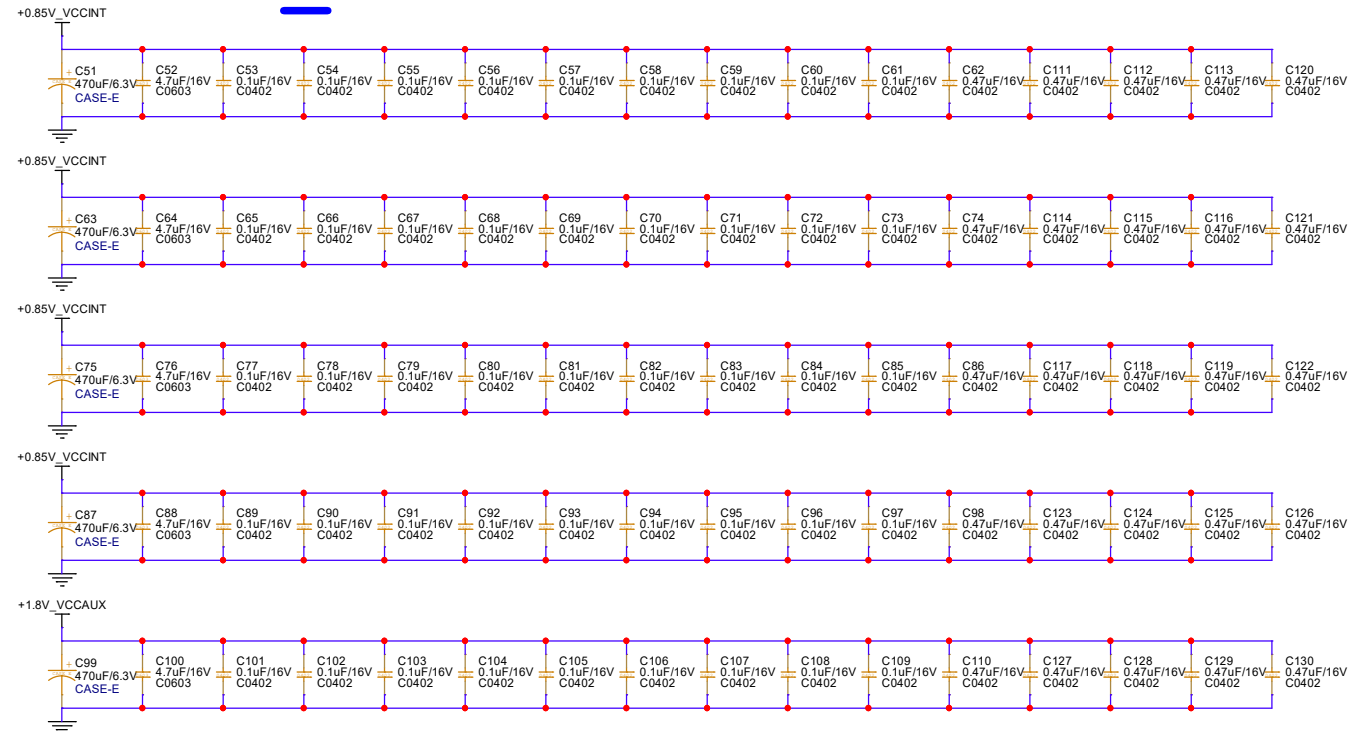
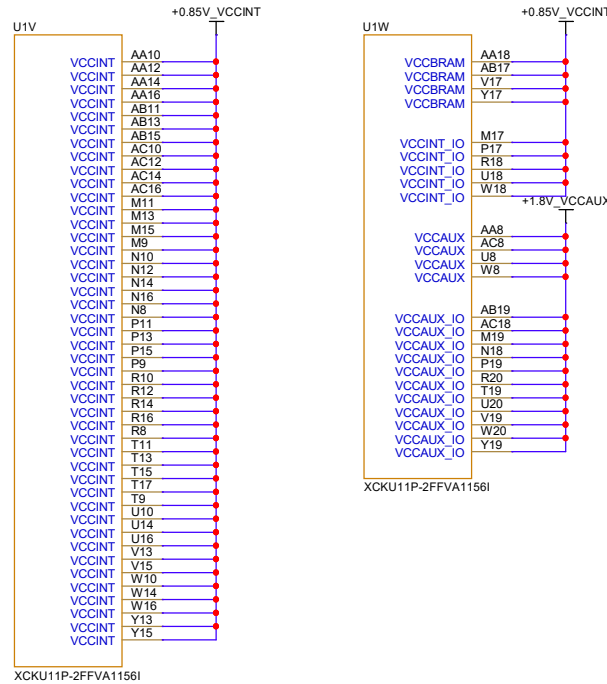
FPGA_PCIE_RXP[0:7]	≪ FPGA_PCIE_RXP[0:7]	[35]
FPGA_PCIE_RXN[0:7]	≪ FPGA_PCIE_RXN[0:7]	[35]
FPGA_PCIE_TXP[0:7]	≪ FPGA_PCIE_TXP[0:7]	[35]
FPGA_PCIE_TXN[0:7]	≪ FPGA_PCIE_TXN[0:7]	[35]
FPGA_PCIE_CLKP	≪ FPGA_PCIE_CLKP	[35]
FPGA_PCIE_CLKN	≪ FPGA_PCIE_CLKN	[35]

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FPGA_GND&NC

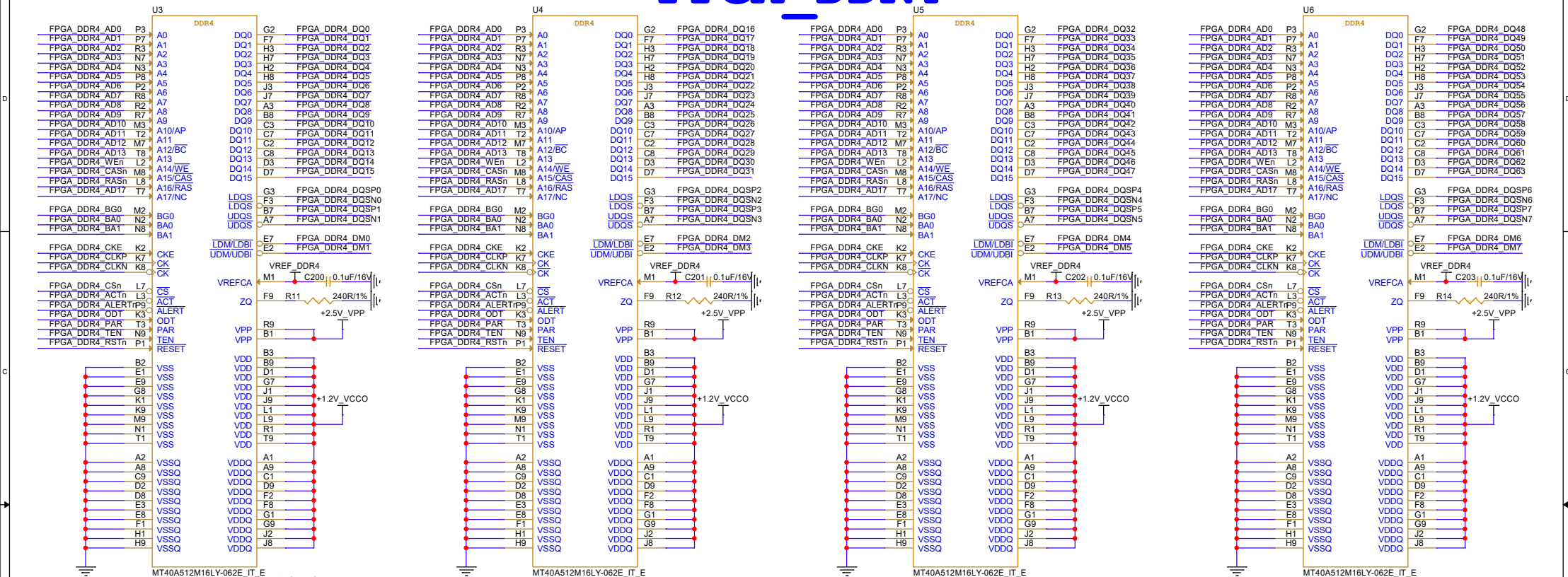


FPGA_PWR



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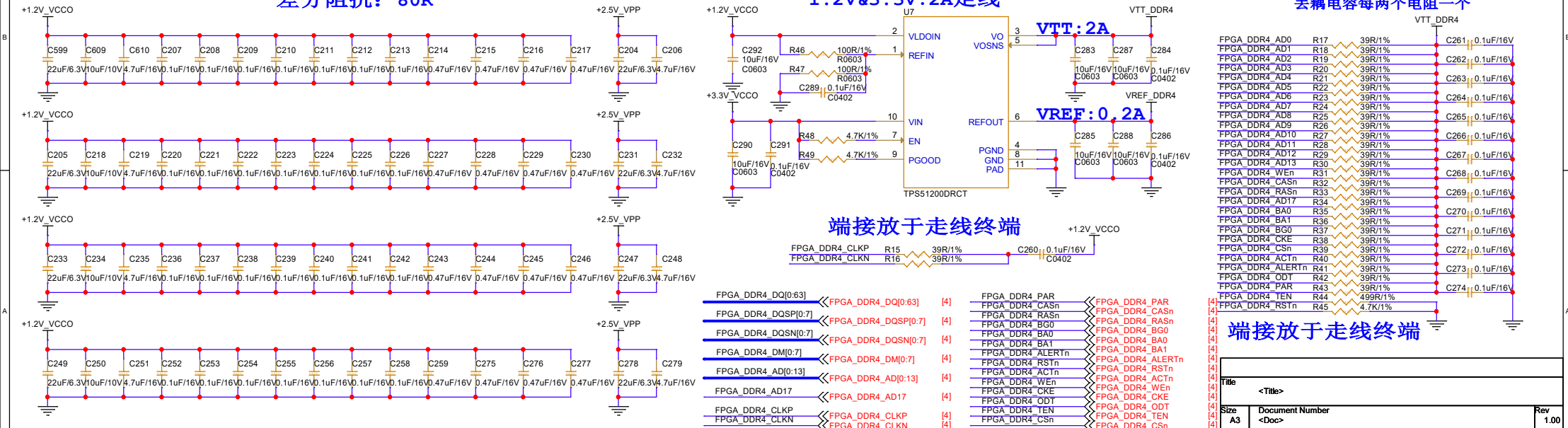
FPGA DDR4



单端阻抗: 40R
差分阻抗: 80R

1.2V&3.3V:2A走线

去耦电容每两个电阻一个



端接放于走线终端

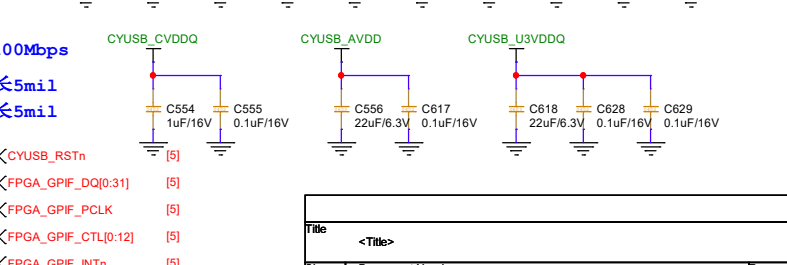
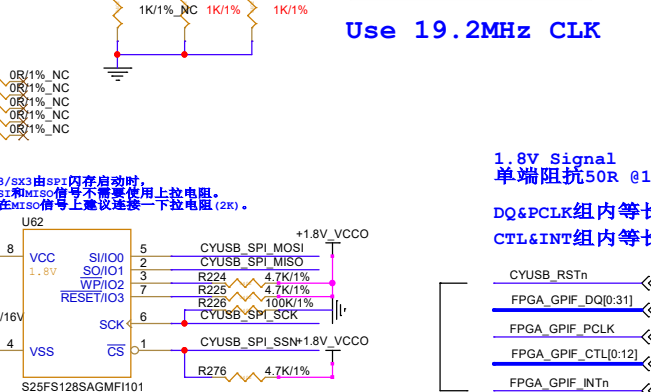
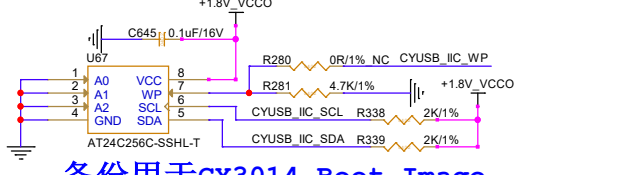
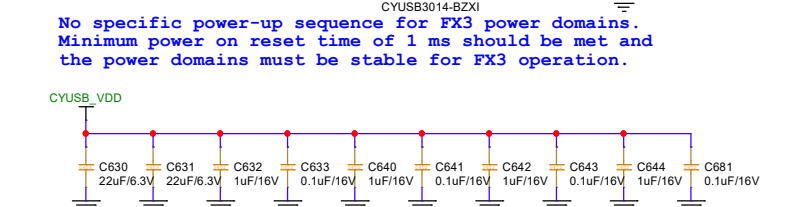
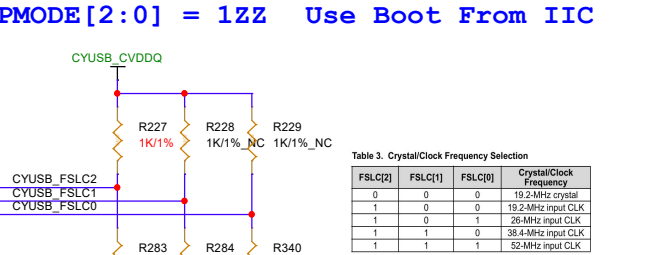
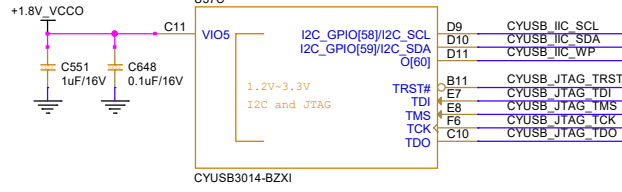
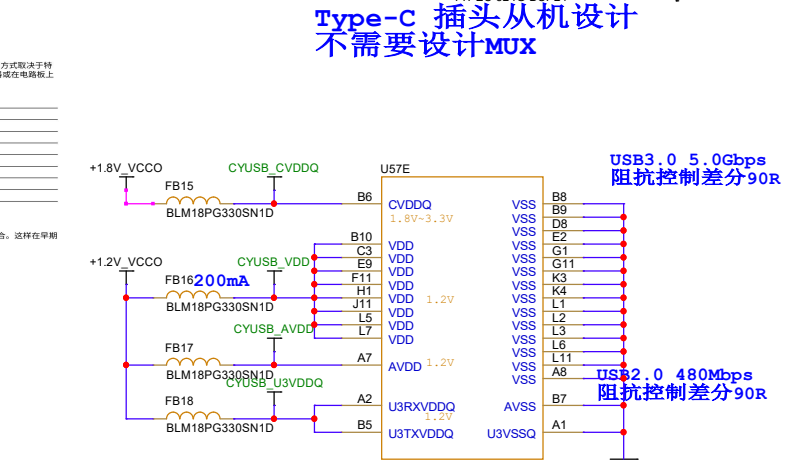
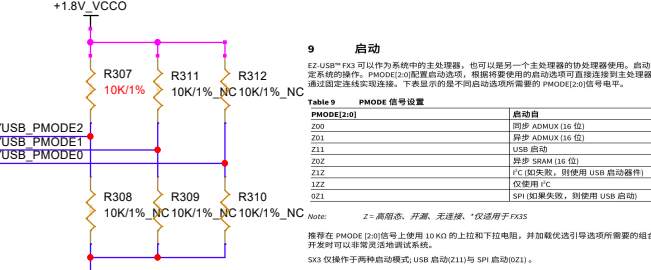
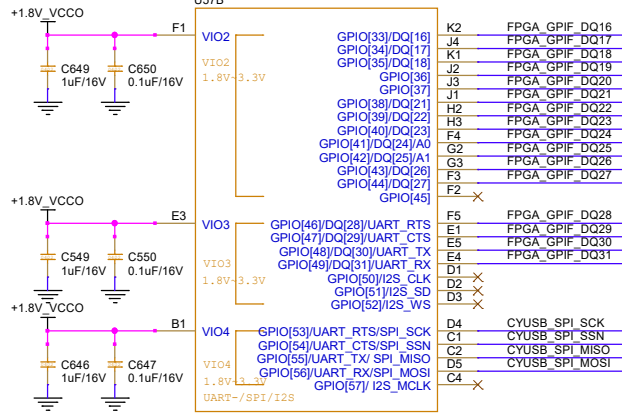
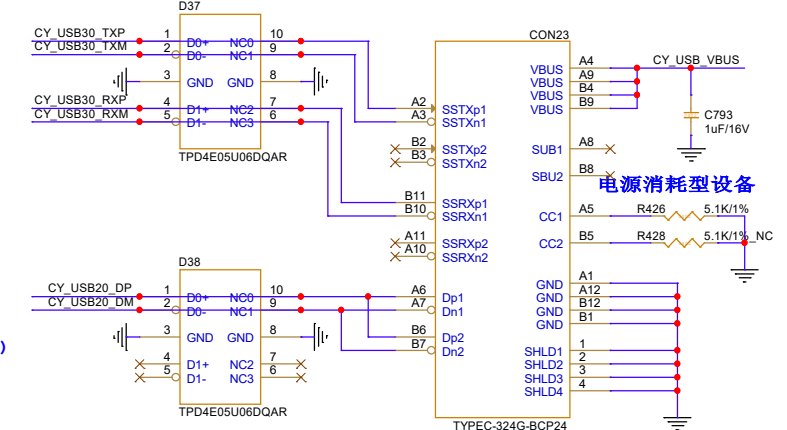
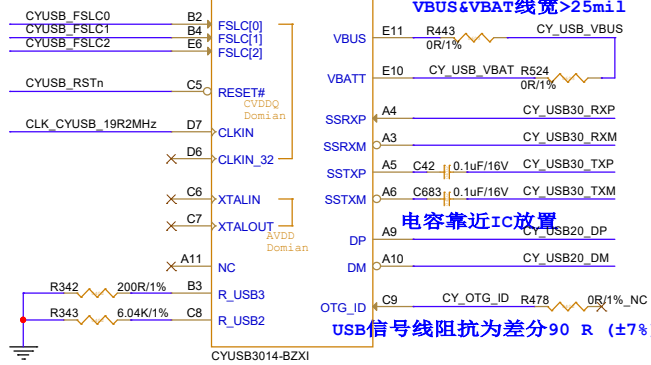
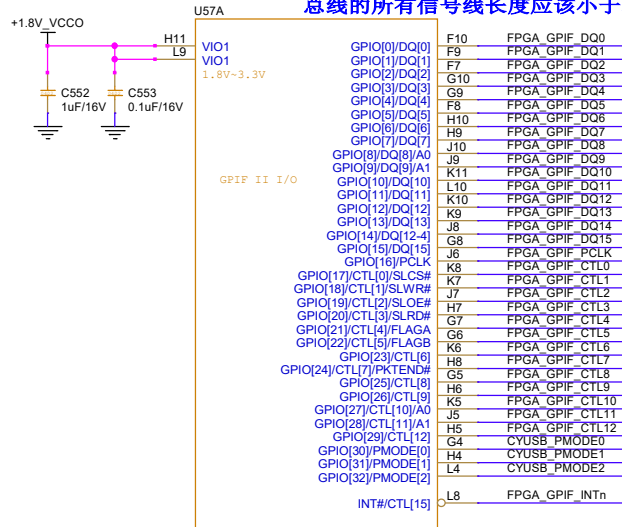
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[4]	A3	<Doc>	1.00
[4]	Date: Friday, July 10, 2026		
[4]	Sheet	11	of 35

USB_PHY

设计参考

infineon-an70707-ez-usb-fx3-fx3s-hardware-design-guidelines-and-schematic-checklist-chinese-applicationnotes-cn.pdf

GPIF II 接口的最大频率是100 MHz。
总线的所有信号线长度应该小于500 mil。

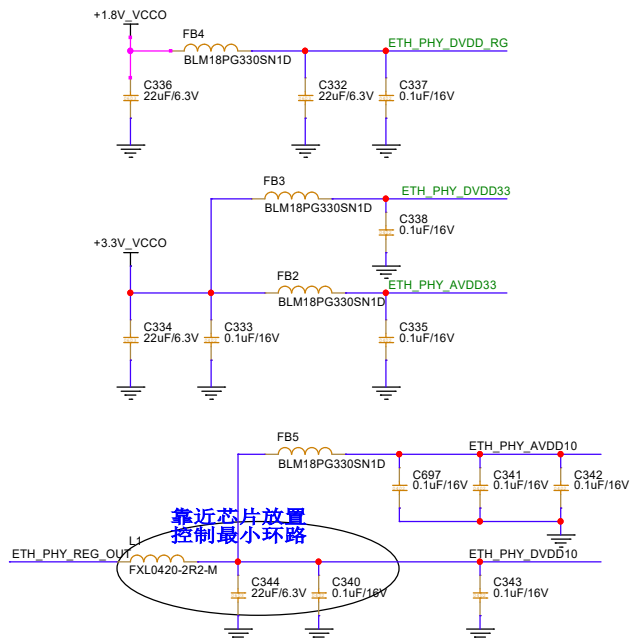
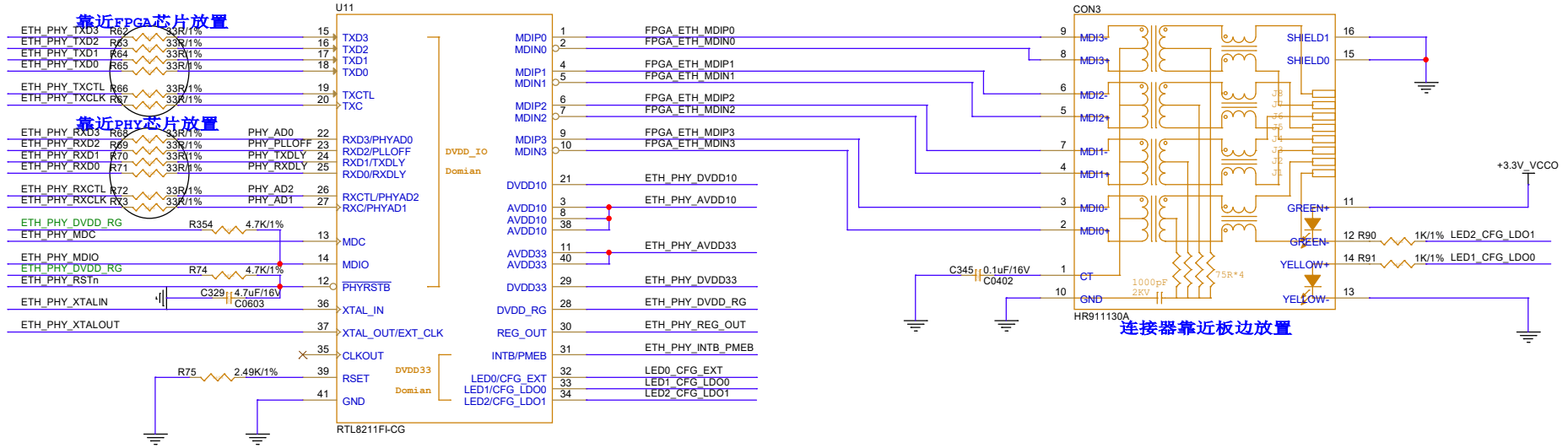


备份用于CY3014 Boot Image

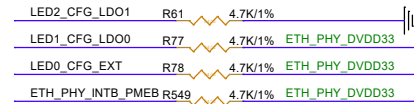
Nor Flash用于CY3014 Boot Image

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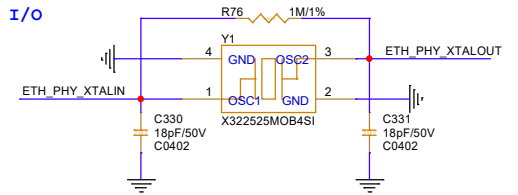
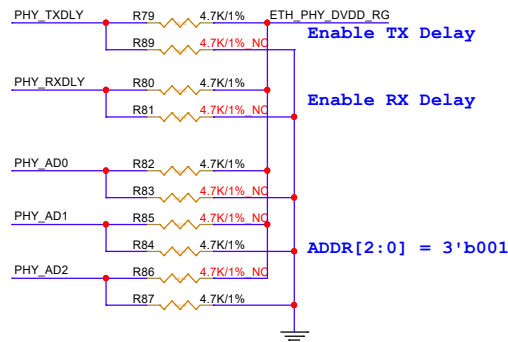
ETH_PHY



RGMII I/O Pad External Power Source Mode Configuration.
1: Use the external power source for the RGMII I/O pad
(via 4.7k-ohm to 3.3V)
0: Use the integrated LDO for the RGMII I/O pad
(via 4.7k-ohm to GND)



V_SEL=10: RGMII IO Voltage is 1.8V
Use external power for RGMII I/O



ETH_PHY_TXD[0:3]	<ETH_PHY_TXD[0:3]	[7]
ETH_PHY_TXCTL	<ETH_PHY_TXCTL	[7]
ETH_PHY_TXCLK	<ETH_PHY_TXCLK	[7]
ETH_PHY_RXD[0:3]	<ETH_PHY_RXD[0:3]	[7]
ETH_PHY_RXCTL	<ETH_PHY_RXCTL	[7]
ETH_PHY_RXCLK	<ETH_PHY_RXCLK	[7]
ETH_PHY_MDC	<ETH_PHY_MDC	[7]
ETH_PHY_MDIO	<ETH_PHY_MDIO	[7]
ETH_PHY_RSTn	<ETH_PHY_RSTn	[7]

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PHY_RSXXX

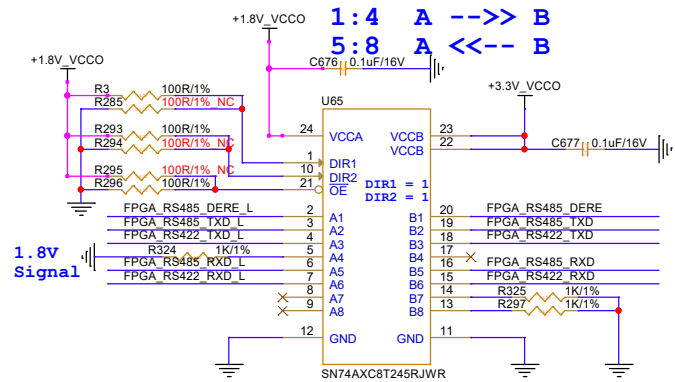
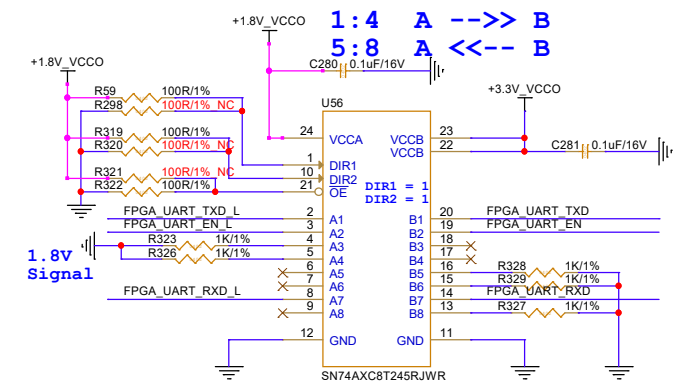


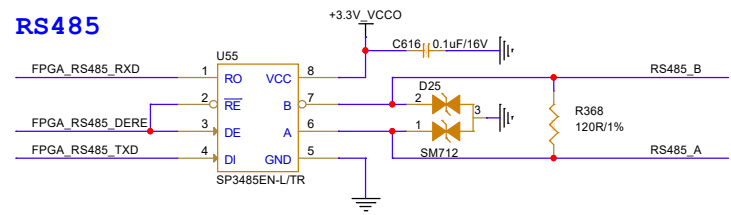
表 7-1. 功能表

控制输入 ⁽¹⁾			信号方向	
OE	DIR1	DIR2	位 1:4	位 5:8
H	X	X	禁用 (高阻态)	
L	L	L	B 转 A	
L	L	H	B 转 A	A 转 B
L	H	L	A 转 B	
L	H	H	A 转 B	B 转 A

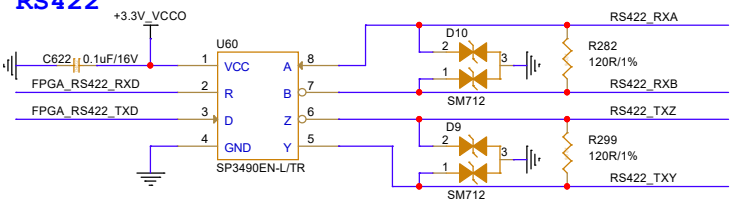
(1) 数据 I/O 的输入电路始终处于激活状态，并必须驱动至有效逻辑电平。



RS485



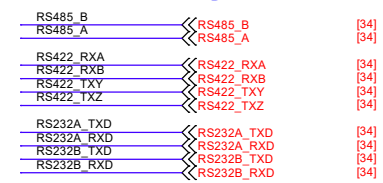
RS422



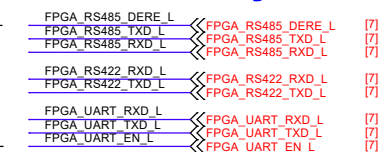
3.3V Level Signal



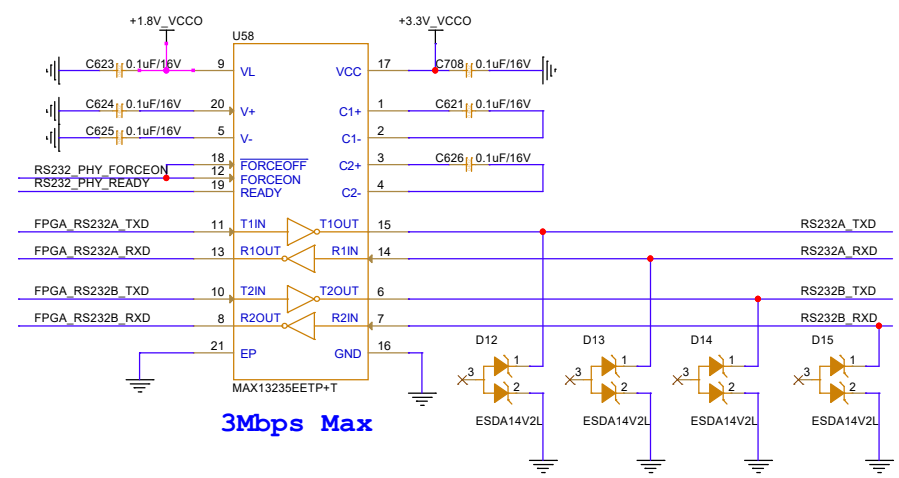
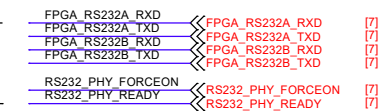
Comm Bus Signal



1.8V Level Signal



1.8V Level Signal



TVS靠近连接器放置

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AD5791BRUZ (20bit, 1Mps, 单通道, 电压输出)
输出电压范围: $\pm 2.5V$, $\pm 5V$, $\pm 10V$ 一共三个范围可调节 (电压输出);
输入接口形式: 2个SMA。 (2个这个芯片)

The schematic diagram illustrates the power and signal connections for the AD5791B evaluation board. The AD5791B IC (U20) is the central component, with its pins connected to various power supplies and signal lines. The power supplies include +1.8V VCCO, +5.0V AVDD, +15V AVCC, and -15V AVEE. The signal lines include DAC outputs (SCLK, SDO, SDIN, SYNC, LDAC, CLR, RESET) and reference voltages (VREFP, VREFN). The AD5791B is connected to two AD8676ARZ comparators (U21A and U21B) which output +5V~+12.5V and -12.5V~0V. The diagram also shows the connection of the AD5791B to the AD5791B evaluation board (U21C).

VCC (5.0V)

DAC_AD5791_1_SYNCn	DAC_AD5791_1_SYNCn	[5]
DAC_AD5791_1_SCLK	DAC_AD5791_1_SCLK	[5]
DAC_AD5791_1_SDO	DAC_AD5791_1_SDO	[5]
DAC_AD5791_1_SDIN	DAC_AD5791_1_SDIN	[5]
DAC_AD5791_1_LDACn	DAC_AD5791_1_LDACn	[5]
DAC_AD5791_1_CLRn	DAC_AD5791_1_CLRn	[5]
DAC_AD5791_1_RSTn	DAC_AD5791_1_RSTn	[5]

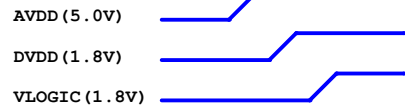
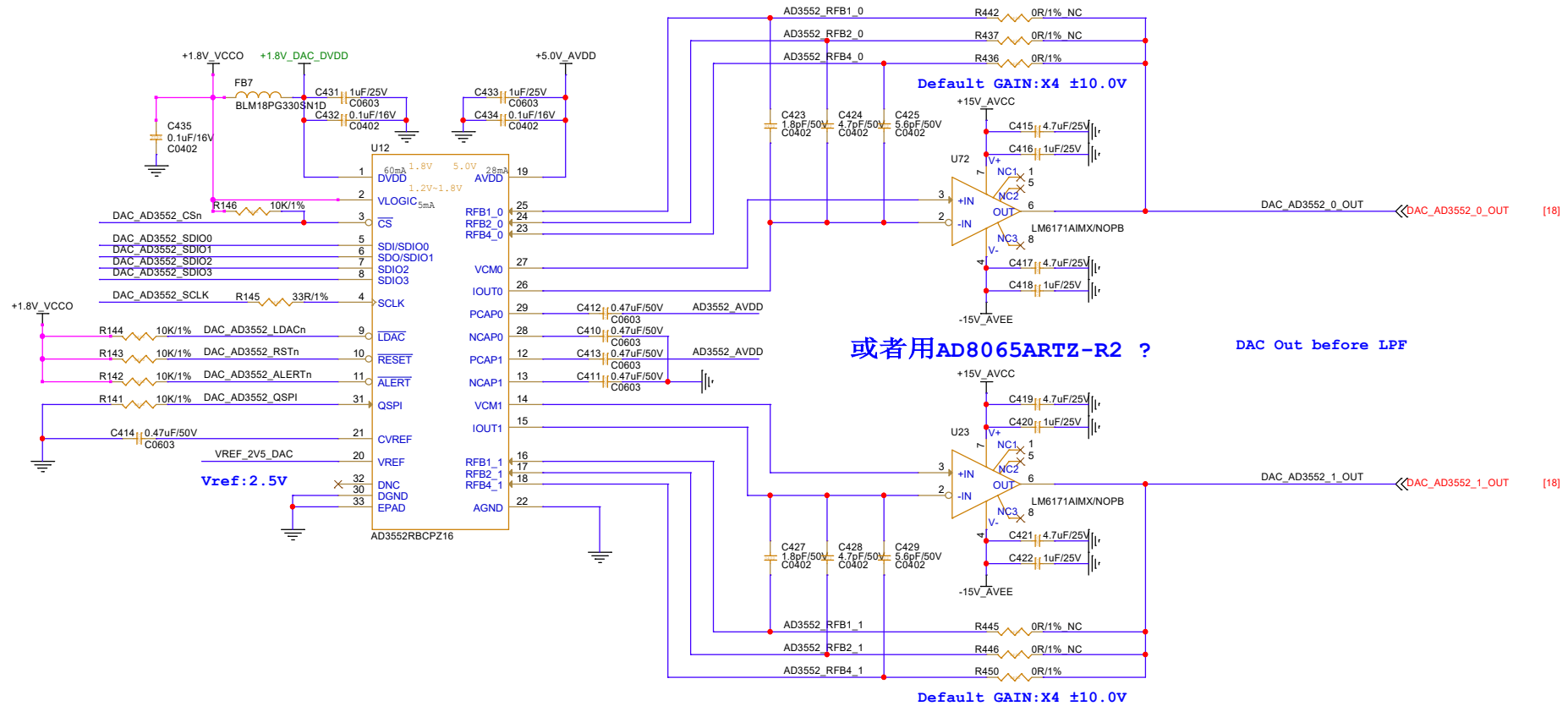
信号线阻抗:
单端50R

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DAC_AD3552

AD3552RBCPZ16-RL7 (16bit, 33Mbps, 双通道, 电流输出)
后端要求转换, 将输出的电流信号转为电压信号;
输出电压范围: $\pm 2.5V$, $\pm 5V$, $\pm 10V$ 一共三个范围可调节 (电压输出);
输出接口形式: 2个SMA。

参考UG-2217 EVAL-AD3552R



POWER SUPPLY RECOMMENDATIONS

The AD3552R does not have any restriction for power supply sequencing. The chip incorporates a power monitor for AV_{DD} and DV_{DD} that releases the internal reset when both rails are within specification. Nevertheless, the recommended sequence to turn on the supply rails is GND, AV_{DD} , DV_{DD} , V_{LOGIC} because it minimizes the power-up glitch.

POWER REQUIREMENTS					
V_{LOGIC} Pin	I_{LOGIC}	1.1	1.8	1.89	V
	V_{LOGIC} Current	1	7.5		μA
	V_{LOGIC} Dynamic Current	3	5		mA
DV_{DD} Pin	I_{DD}	1.71	1.8	1.89	V
	DV_{DD} Current		0.5	0.8	mA
	DV_{DD} Dynamic Current		53	60	mA
AV_{DD} Pin	I_{DD}	4.75	5	5.25	V
	AV_{DD} Current		22	28.5	mA
	AV_{DD} Power-Down Current		0.6		mA
AV_{DD} Reset Current	I_{DD}		120		μA

$V_{IH} = V_{LOGIC} \times 0.9$, $V_{IL} = V_{LOGIC} \times 0.1$
 $SCLK = 66$ MHz, quad SPI DDR, $V_{IH} = V_{LOGIC} \times 0.85$, $V_{IL} = V_{LOGIC} \times 0.35$
 $SCLK = 66$ MHz, quad SPI DDR, simultaneous update
Channel 0 and Channel 1 zero-scale, 0 V to ± 5 V range
After reset, DACs powered down
RESET asserted

Vref信号20mil宽走线
双边地线伴随走线

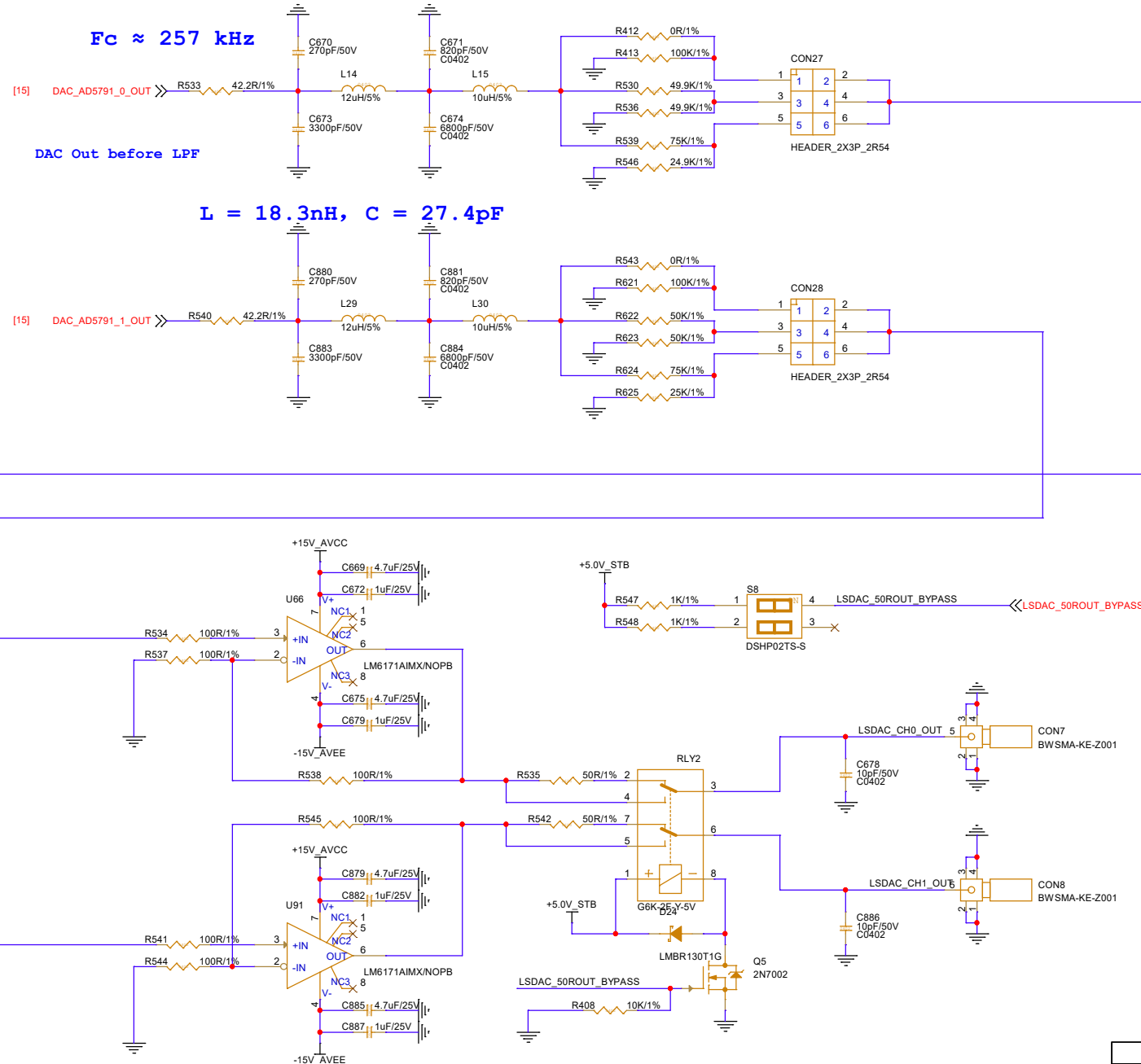
VREF_2V5_DAC	<<VREF_2V5_DAC	[32]
DAC_AD3552_SDIO[0:3]	<<DAC_AD3552_SDIO[0:3]	[5]
DAC_AD3552_SCLK	<<DAC_AD3552_SCLK	[5]
DAC_AD3552_LDACn	<<DAC_AD3552_LDACn	[5]
DAC_AD3552_RSTn	<<DAC_AD3552_RSTn	[5]
DAC_AD3552_ALERTn	<<DAC_AD3552_ALERTn	[5]
DAC_AD3552_QSPI	<<DAC_AD3552_QSPI	[5]
DAC_AD3552_CSn	<<DAC_AD3552_CSn	[5]

单端阻抗: 50R

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LSDAC_LPF

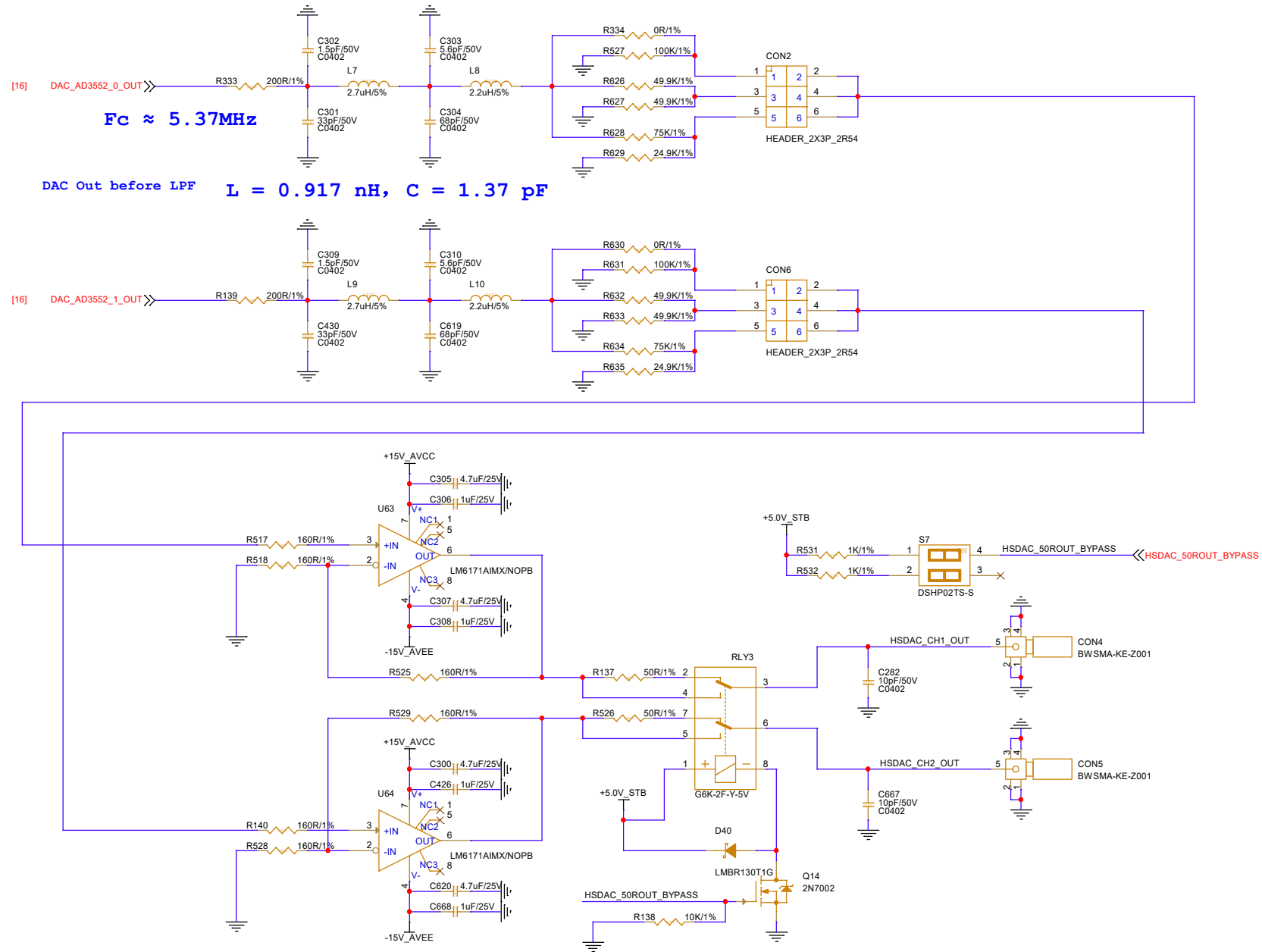
低速DAC的滤波截止频率设计为250KHZ
采用二阶Bessel低通滤波



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HSDAC_LPF

高速DAC的滤波截止频率设计为5MHz
采用二阶Bessel低通滤波



VADJ AD4630

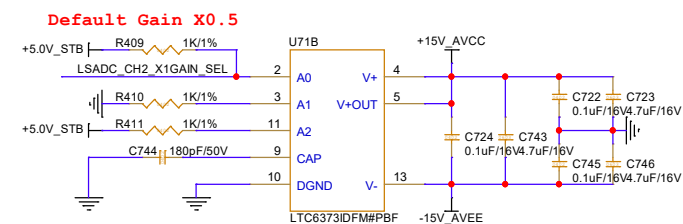
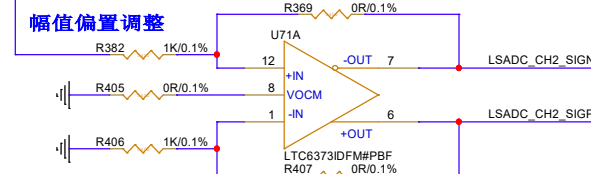
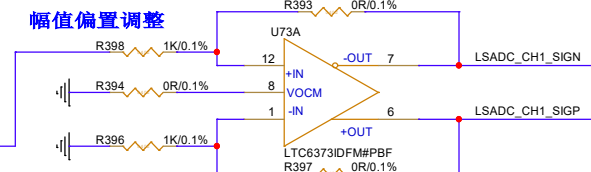
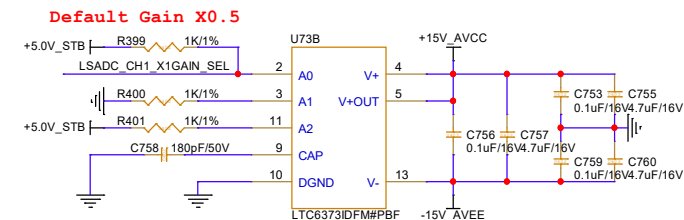
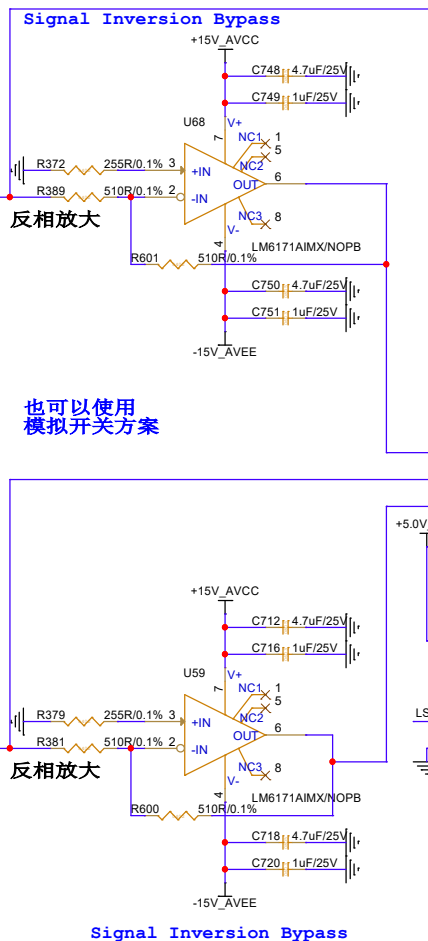
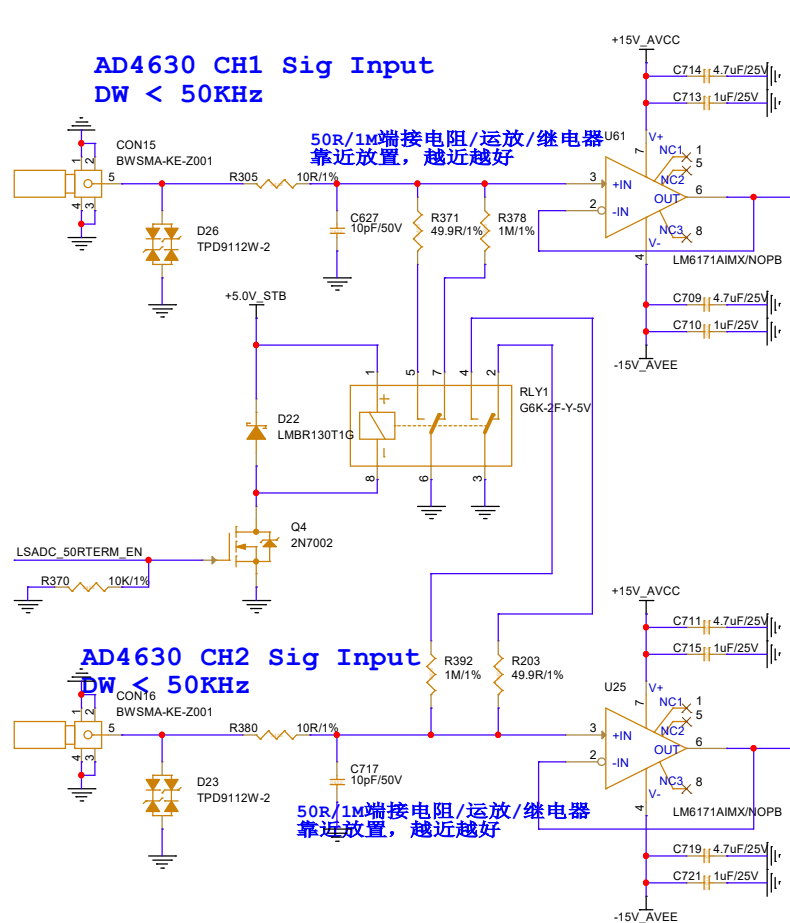
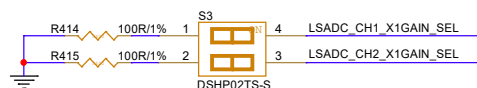
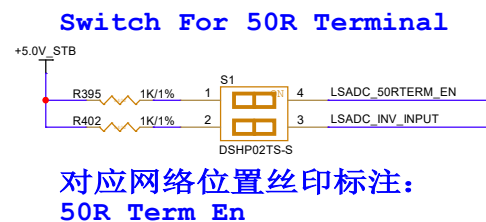


Table 1. Gain Selection Table for LTC637

A2	A1	A0	G = GAIN SETTING (V/V)
L	L	L	16
L	L	H	8
L	H	L	4
L	H	H	2
H	L	L	1
H	L	H	0.5
H	H	L	0.25
H	H	H	Shutdown

Switch For Input Range Toggle

Switch-ON For 0~+5V Input Uses
Default use for 0V~+10V input usage

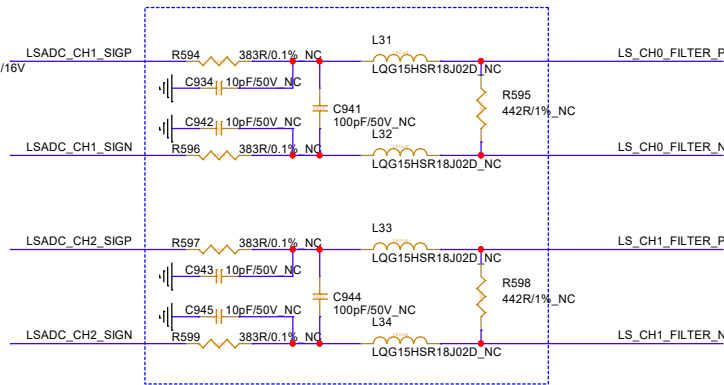
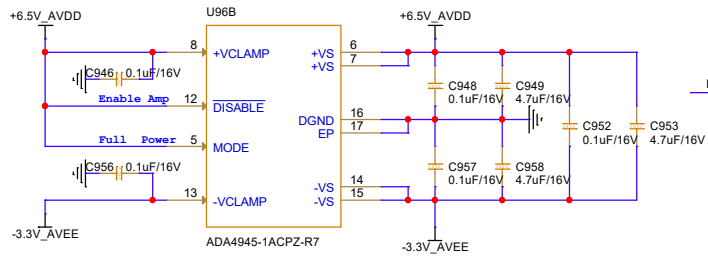
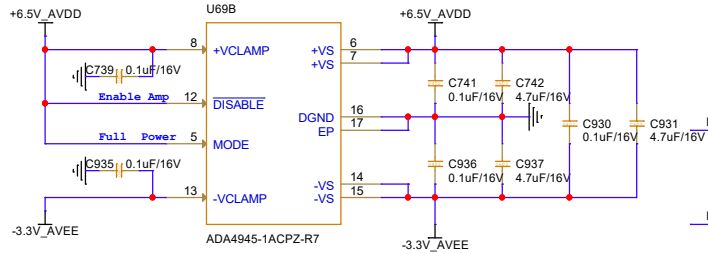
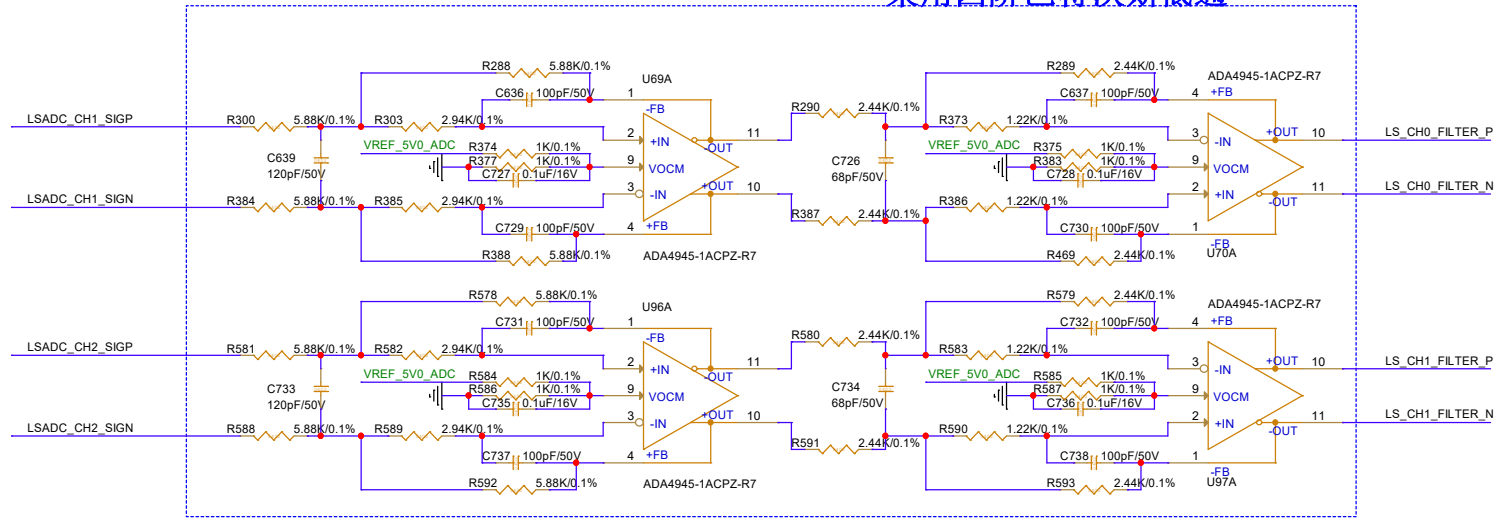


LSADC_CH1_SIGP	↔	LSADC_CH1_SIGP	[20]
LSADC_CH1_SIGN	↔	LSADC_CH1_SIGN	[20]
LSADC_CH2_SIGP	↔	LSADC_CH2_SIGP	[20]
LSADC_CH2_SIGN	↔	LSADC_CH2_SIGN	[20]
VREF_5V0_ADC	↔	VREF_5V0_ADC	[20,21,32]

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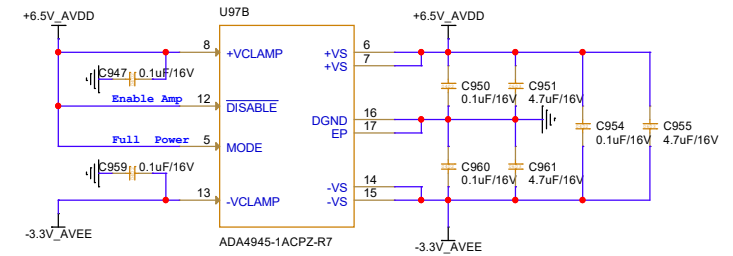
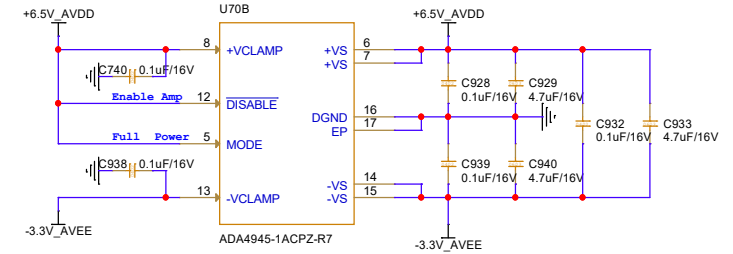
AD4630_LPF

高速ADC截至频率设计为250KHZ
采用四阶巴特沃斯低通



Backup Passive LPF

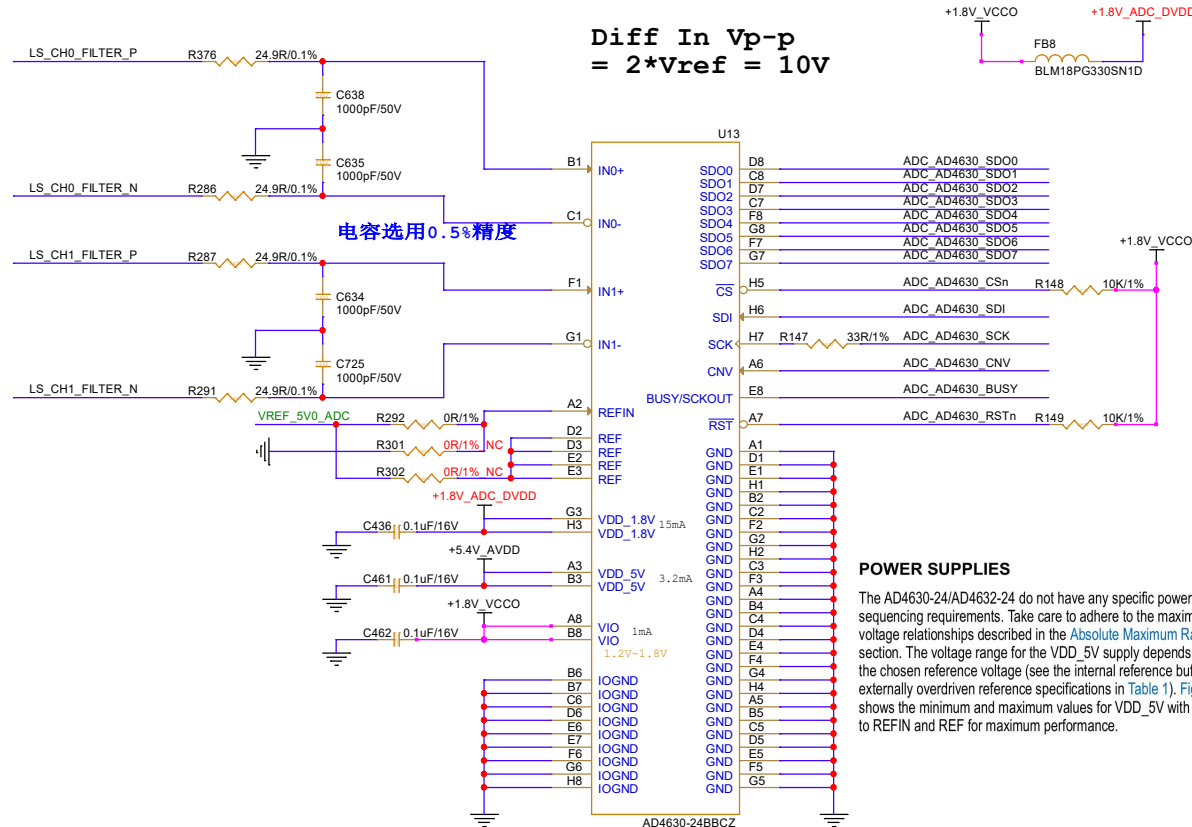
LSADC_CH1_SIGP << LSADC_CH1_SIGP [19]
LSADC_CH1_SIGN << LSADC_CH1_SIGN [19]
LSADC_CH2_SIGP << LSADC_CH2_SIGP [19]
LSADC_CH2_SIGN << LSADC_CH2_SIGN [19]
LS_CH0_FILTER_P << LS_CH0_FILTER_P [21]
LS_CH0_FILTER_N << LS_CH0_FILTER_N [21]
LS_CH1_FILTER_P << LS_CH1_FILTER_P [21]
LS_CH1_FILTER_N << LS_CH1_FILTER_N [21]
VREF_5V0_ADC << VREF_5V0_ADC [21,32]



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ADC_AD4630

AD4630-24BBCZ (24bit, 2MSPS, 双通道, 差分输入)
输入电压范围: 0-5V, 0-10V, 0- -10V, 三个范围可调节 (单端输入);
输入接口形式: 2个SMA。 (1个这个芯片);
增加50欧和1M欧的不同输入阻抗调节
(信号最高频率30MHz):



VDD_5V:
5.3 V to 5.5 V for a 5 V reference,
4.75 V to 5.25 V for a 4.096 V reference

Analog Signal

LS_CH0_FILTER_P	LS_CH0_FILTER_P	[20]
LS_CH0_FILTER_N	LS_CH0_FILTER_N	[20]
LS_CH1_FILTER_P	LS_CH1_FILTER_P	[20]
LS_CH1_FILTER_N	LS_CH1_FILTER_N	[20]
VREF_5V0_ADC	VREF_5V0_ADC	[20,32]

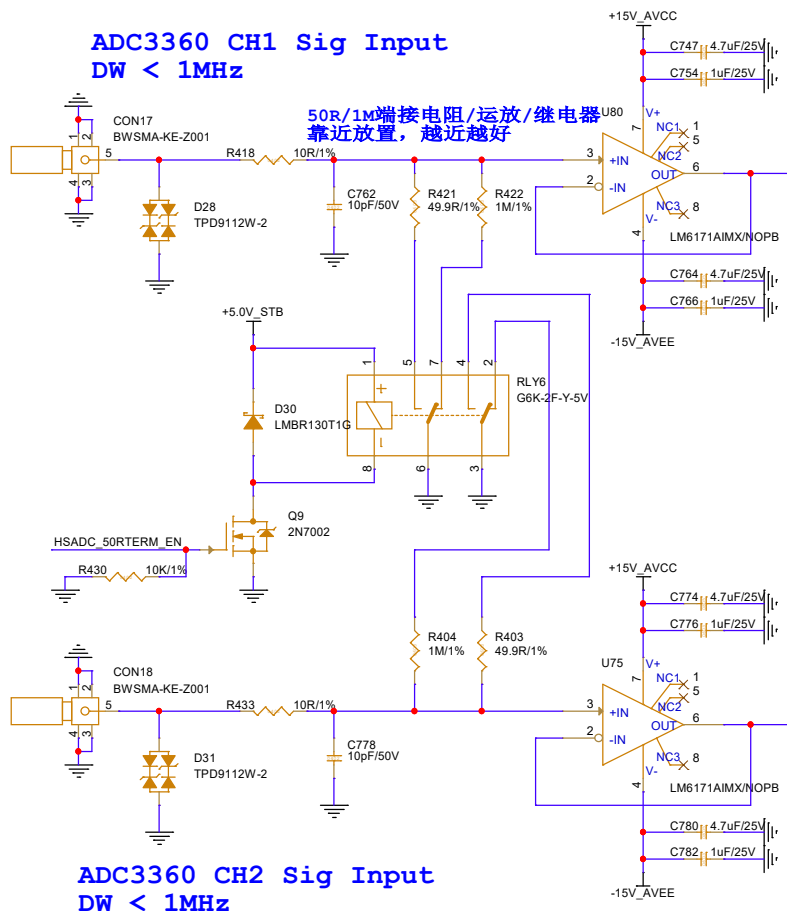
Digital Signal

ADC_AD4630_SDO[0:7]	ADC_AD4630_SDO[0:7]	[5]
ADC_AD4630_CSn	ADC_AD4630_CSn	[5]
ADC_AD4630_SDI	ADC_AD4630_SDI	[5]
ADC_AD4630_SCK	ADC_AD4630_SCK	[5]
ADC_AD4630_CNv	ADC_AD4630_CNv	[5]
ADC_AD4630_BUSY	ADC_AD4630_BUSY	[5]
ADC_AD4630_RSTn	ADC_AD4630_RSTn	[5]

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VADJ_ADC3360

ADC3360 CH1 Sig Input DW < 1MHz



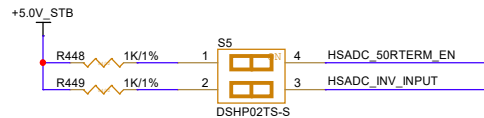
ADC3360 CH2 Sig Input DW < 1MHz

50R/1M端接电阻/运放/继电器
靠近放置，越近越好

Switch For Input Range Toggle

Switch-ON For 0~+5V Input Uses
Default use for 0V~+10V input usage

Switch For 50R Terminal



对应网络位置丝印标注:
50R Term En



网络旁标注:
-10V~0V Input Enable

网络旁标注:
0V~+5V Input Enable

Default Gain X0.25

幅值偏置调整

幅值偏置调整

Default Gain X0.25

Table 1. Gain Selection Table for LTC6373

A2	A1	A0	G = GAIN SETTING (V/V)
L	L	L	16
L	L	H	8
L	H	L	4
L	H	H	2
H	L	L	1
H	L	H	0.5
H	H	L	0.25
H	H	H	Shutdown

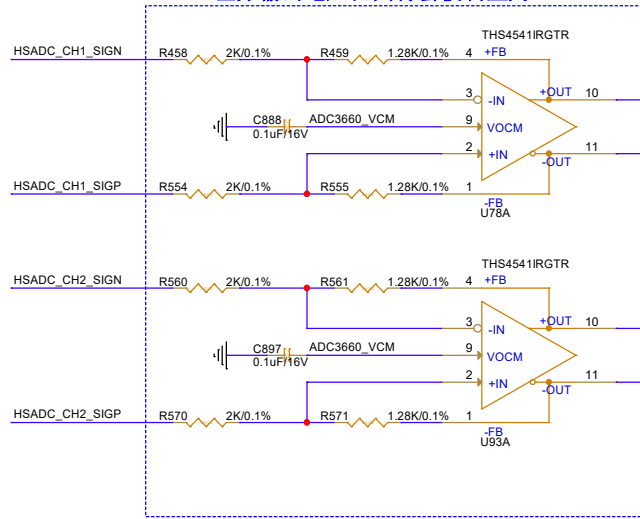
HSADC_CH1_SIGP	HSADC_CH1_SIGP	[23]
HSADC_CH1_SIGN	HSADC_CH1_SIGN	[23]
HSADC_CH2_SIGP	HSADC_CH2_SIGP	[23]
HSADC_CH2_SIGN	HSADC_CH2_SIGN	[23]

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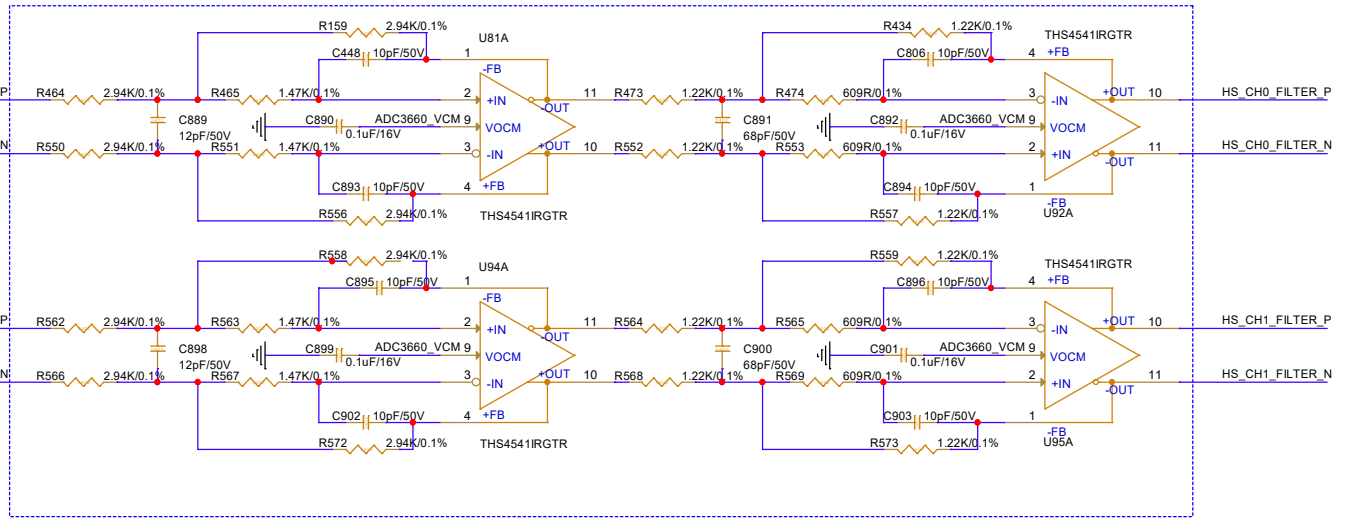
ADC3660_LPF

全差分漏斗放大器级

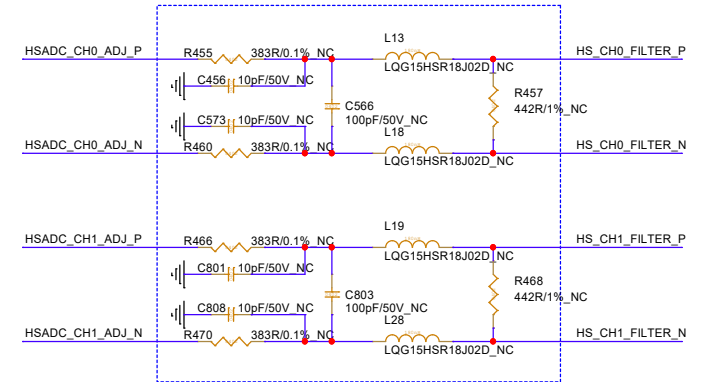
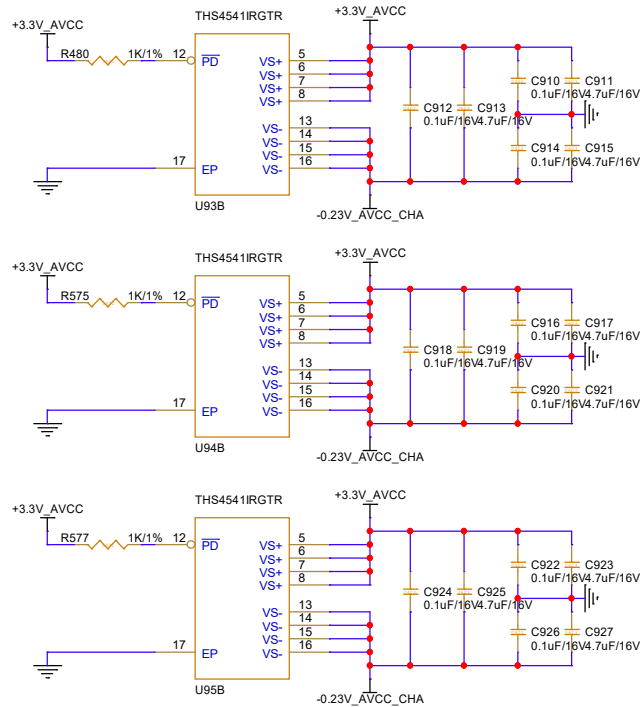
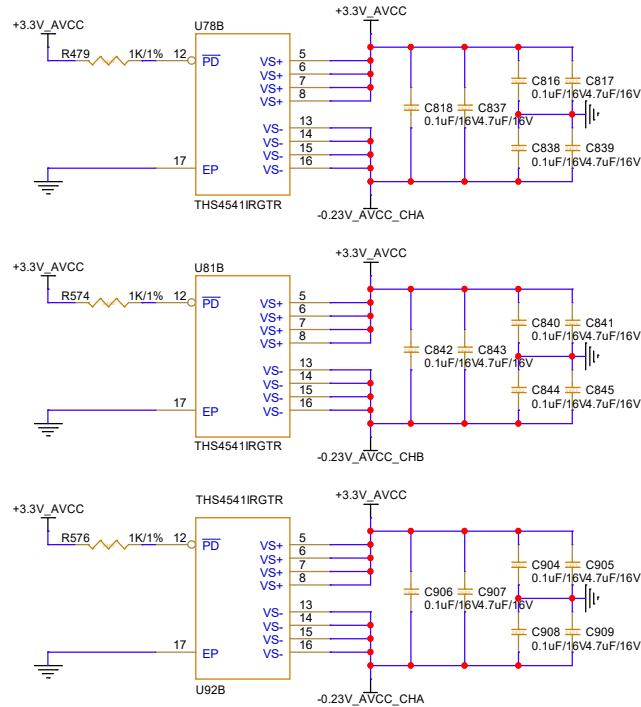
差分输入电压范围需要已调整为 $\pm 2.5V$
差分输出电压范围将会被调整为 $\pm 1.6V$



高速ADC截至频率设计为5MHZ 采用四阶巴特沃斯低通



THS4541IRGTR 与 ADA4932-1
(VQFN-16) 引脚兼容



Backup Passive LPF

ADC3660_VCM	ADC3660_VCM	[24]
HSADC_CH1_SIGP	HSADC_CH1_SIGP	[22]
HSADC_CH1_SIGN	HSADC_CH1_SIGN	[22]
HSADC_CH2_SIGP	HSADC_CH2_SIGP	[22]
HSADC_CH2_SIGN	HSADC_CH2_SIGN	[22]
HS_CH0_FILTER_P	HS_CH0_FILTER_P	[24]
HS_CH0_FILTER_N	HS_CH0_FILTER_N	[24]
HS_CH1_FILTER_P	HS_CH1_FILTER_P	[24]
HS_CH1_FILTER_N	HS_CH1_FILTER_N	[24]

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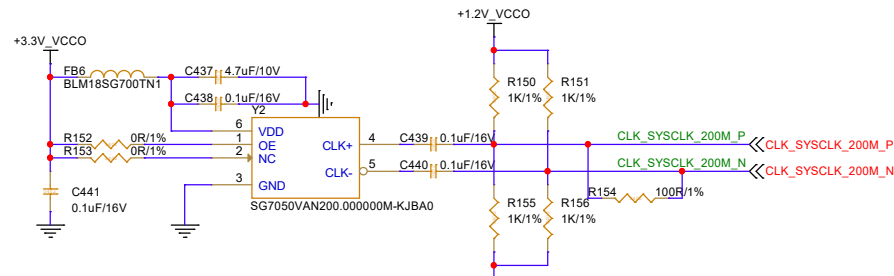
ADC3660IRSBT (16bit, 30Mps, 双通道, 差分输入)
输入电压范围: 0-5V, 0-10V, 0- -10V, 一共三个范围可调节(单端输入);
输入接口形式: 2个SMA。(1个这个芯片);
增加50欧和1M欧的不同输入阻抗调节



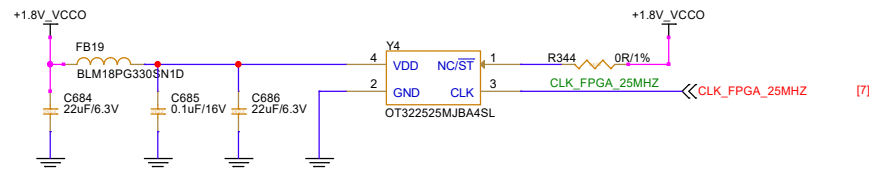
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<u>ADC_ADC3660_CLKN</u>	↔	<u>ADC_ADC3660_CLKN</u>	[5]
<u>ADC_ADC3660_DCLKIN</u>	↔	<u>ADC_ADC3660_DCLKIN</u>	[5]
<u>ADC_ADC3660_DA6</u>	↔	<u>ADC_ADC3660_DA6</u>	[5]
<u>ADC_ADC3660_DA5</u>	↔	<u>ADC_ADC3660_DA5</u>	[5]
<u>ADC_ADC3660_DB6</u>	↔	<u>ADC_ADC3660_DB6</u>	[5]
<u>ADC_ADC3660_DB5</u>	↔	<u>ADC_ADC3660_DB5</u>	[5]
<u>ADC_ADC3660_DCLK</u>	↔	<u>ADC_ADC3660_DCLK</u>	[5]
<u>ADC_ADC3660_FCLK</u>	↔	<u>ADC_ADC3660_FCLK</u>	[5]
<u>ADC_ADC3660_SYNC</u>	↔	<u>ADC_ADC3660_SYNC</u>	[5]
<u>ADC_ADC3660_SCLK</u>	↔	<u>ADC_ADC3660_SCLK</u>	[5]
<u>ADC_ADC3660_SDIO</u>	↔	<u>ADC_ADC3660_SDIO</u>	[5]
<u>ADC_ADC3660_SEN</u>	↔	<u>ADC_ADC3660_SEN</u>	[5]
<u>ADC_ADC3660_RST</u>	↔	<u>ADC_ADC3660_RST</u>	[5]

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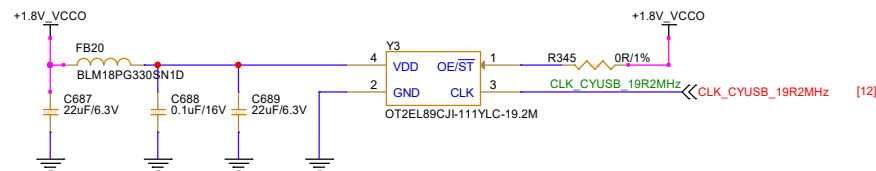
CLK_OSC



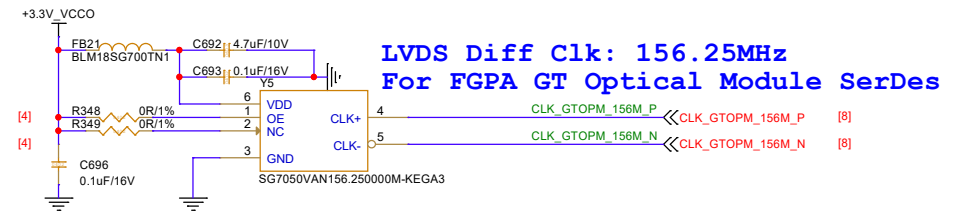
HTSL Diff Clk: 200MHz/300MHz
For FPGA DDRMC



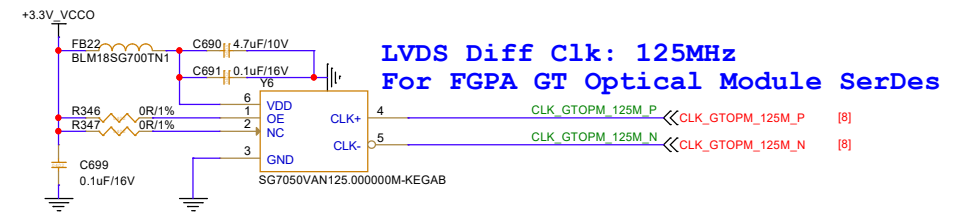
LVCMOS18 Single Clk: 25MHz
For FPGA User Logic



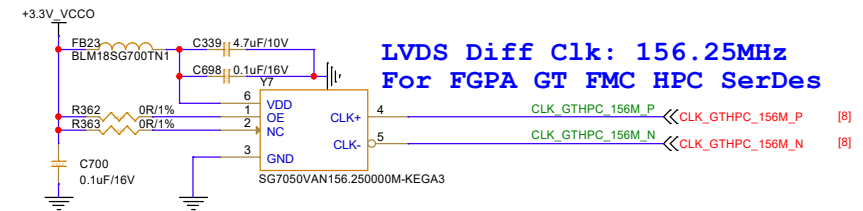
LVCMOS18 Single Clk: 19.2MHz
For CYUSB3014 PHY



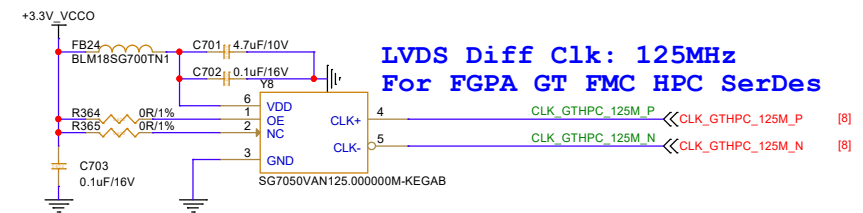
LVDS Diff Clk: 156.25MHz
For FPGA GT Optical Module SerDes



LVDS Diff Clk: 125MHz
For FPGA GT Optical Module SerDes



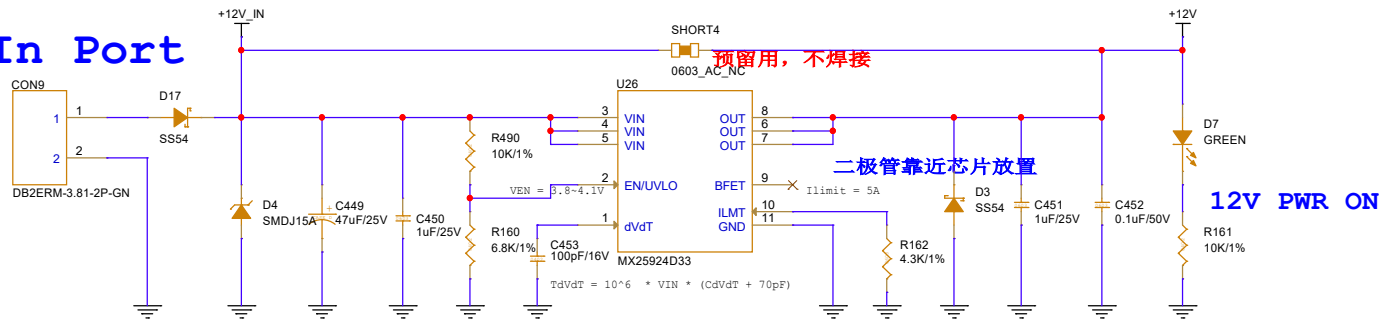
LVDS Diff Clk: 156.25MHz
For FPGA GT FMC HPC SerDes



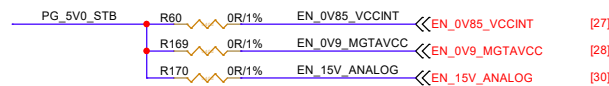
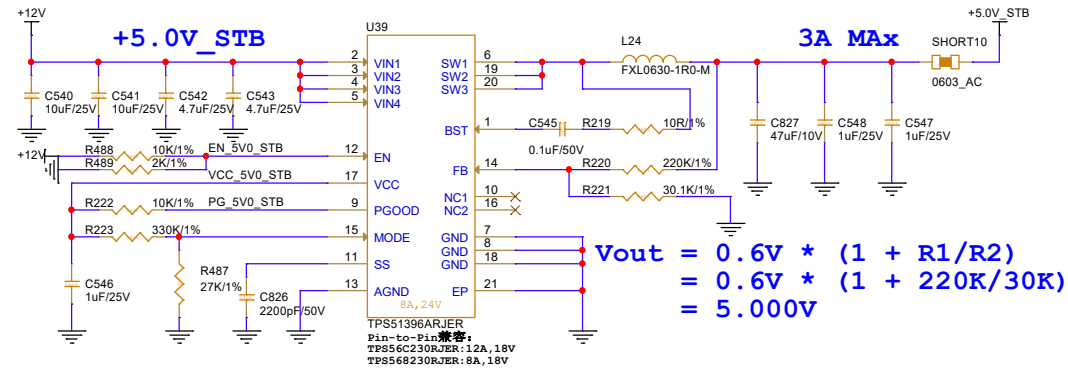
LVDS Diff Clk: 125MHz
For FPGA GT FMC HPC SerDes

PMIC_VIN

Power In Port

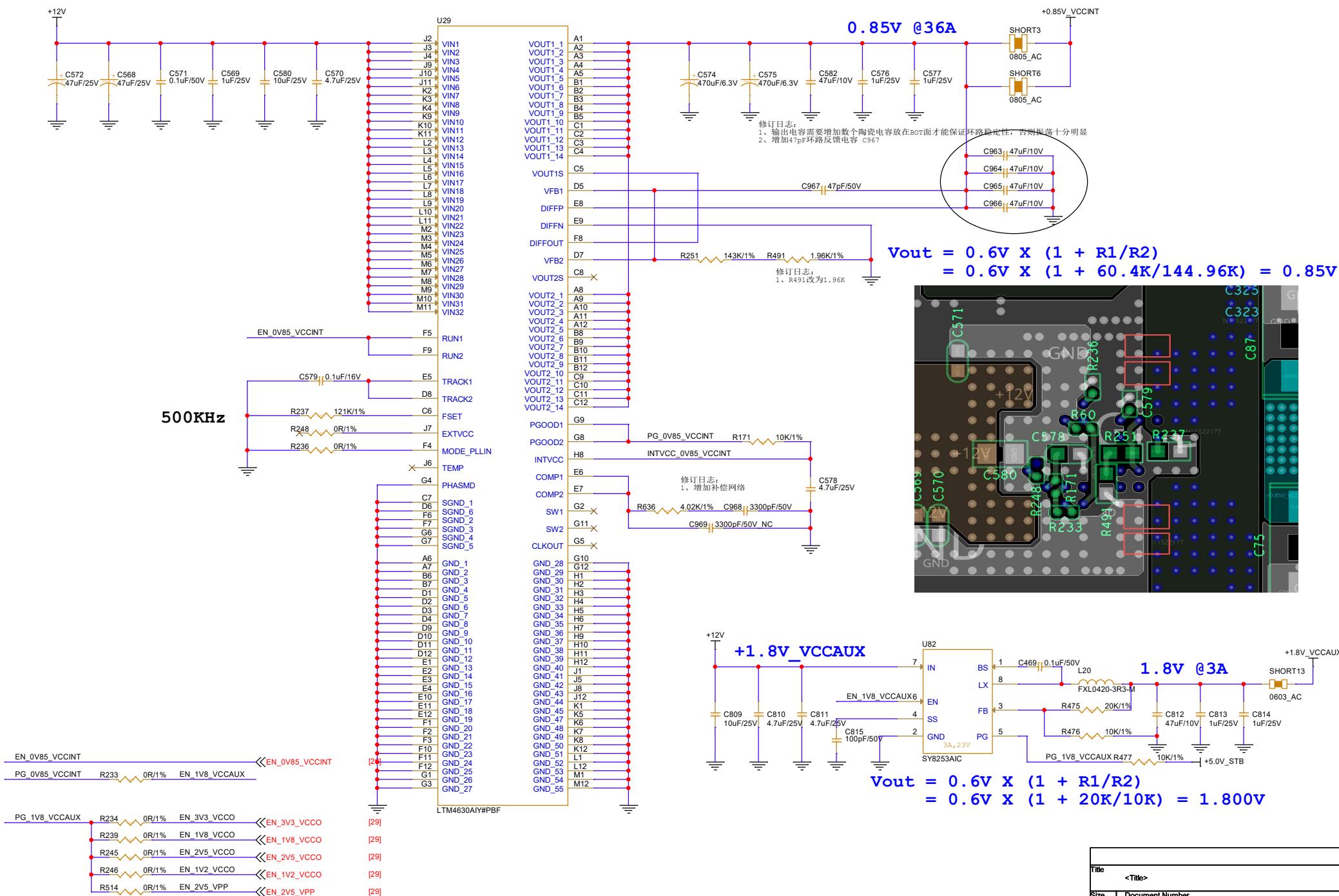


12V直接生成+5.0V_STB, 然后再进行下级电源控制



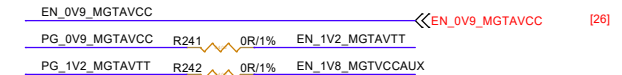
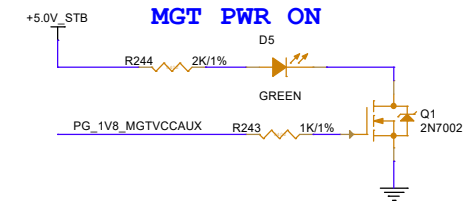
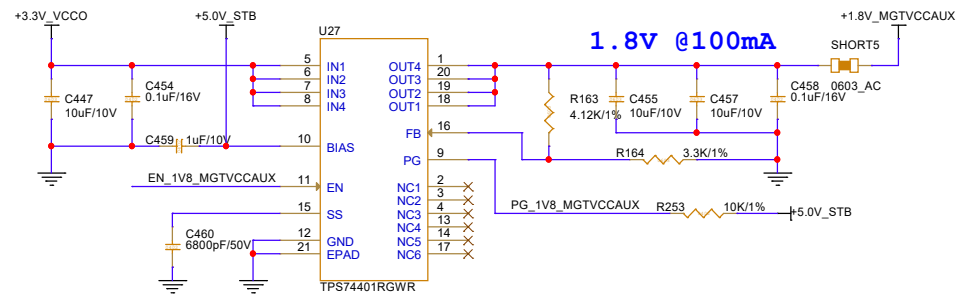
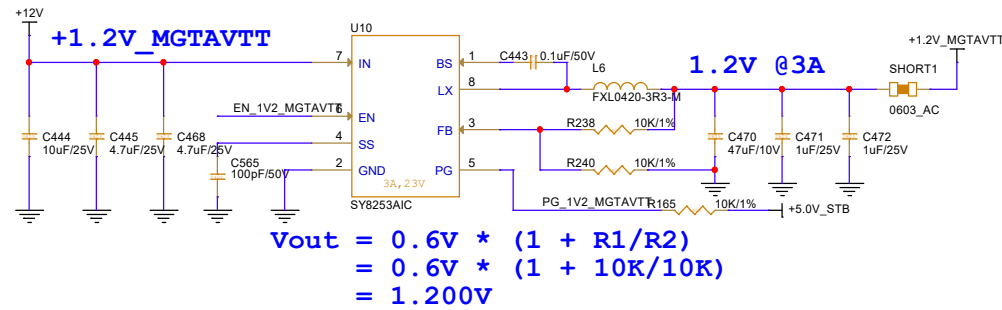
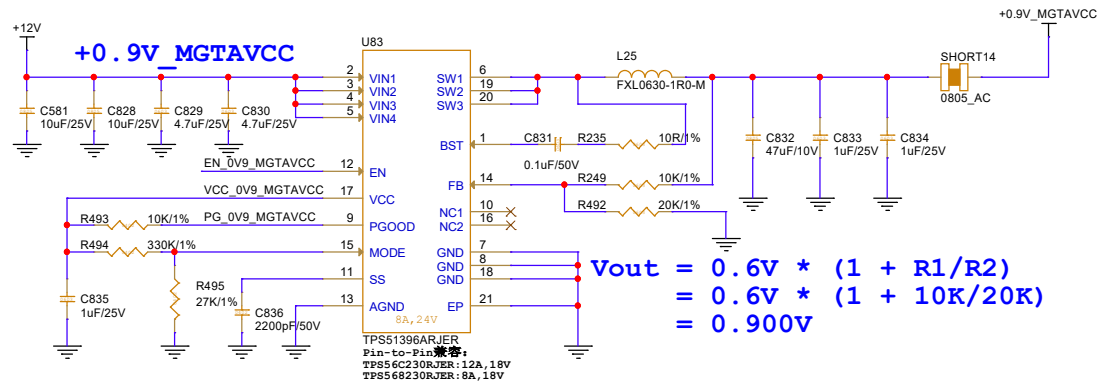
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PMIC_VCCINT&AUX

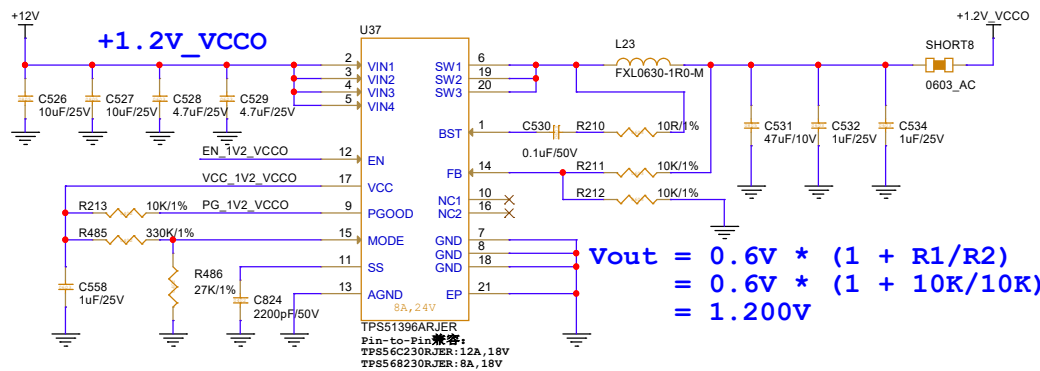
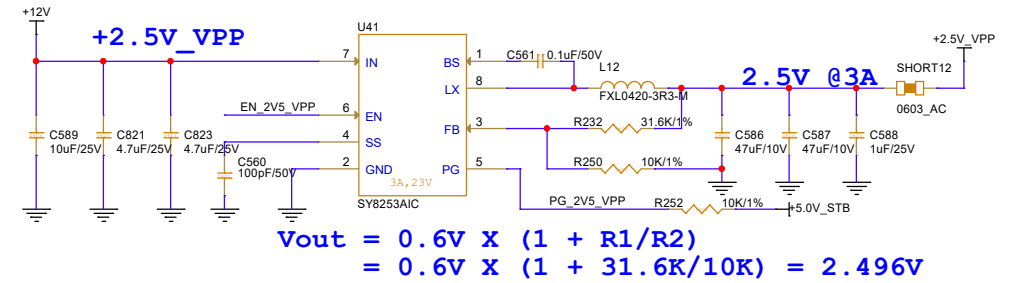
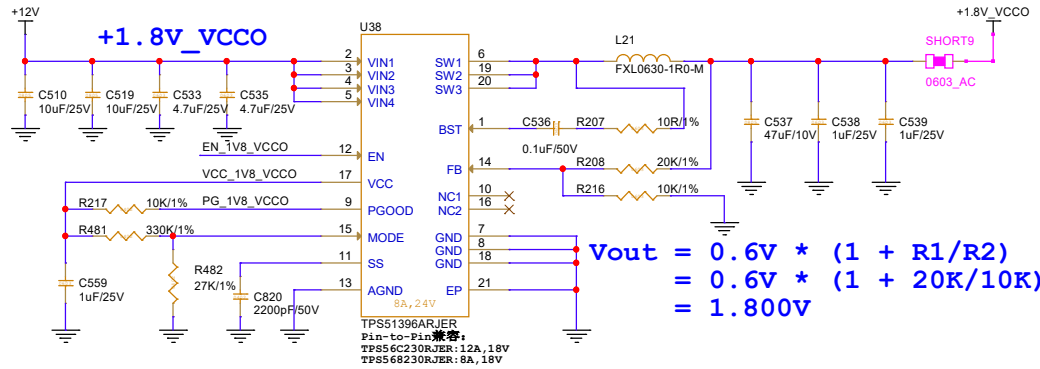
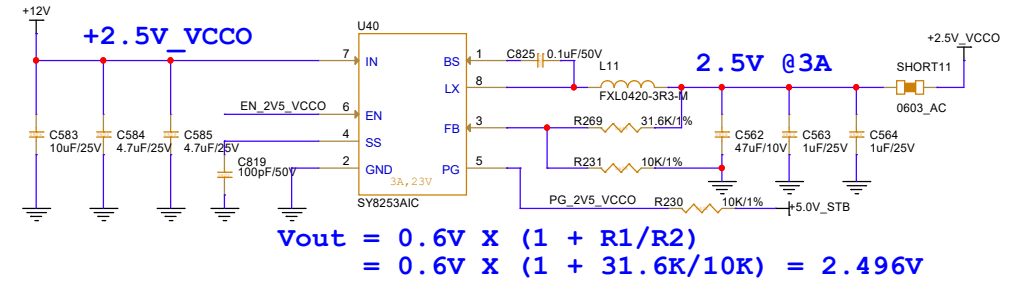
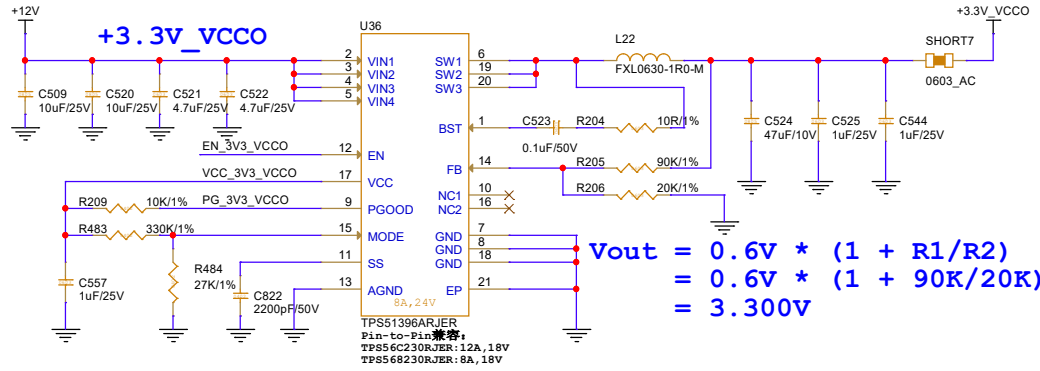


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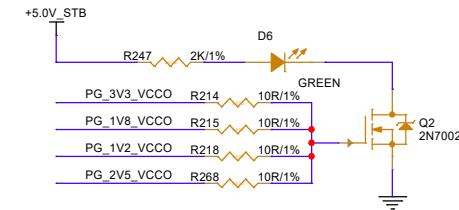
PMIC_FGPA_MGT



PMIC_VCCO



VCCO PWR ON

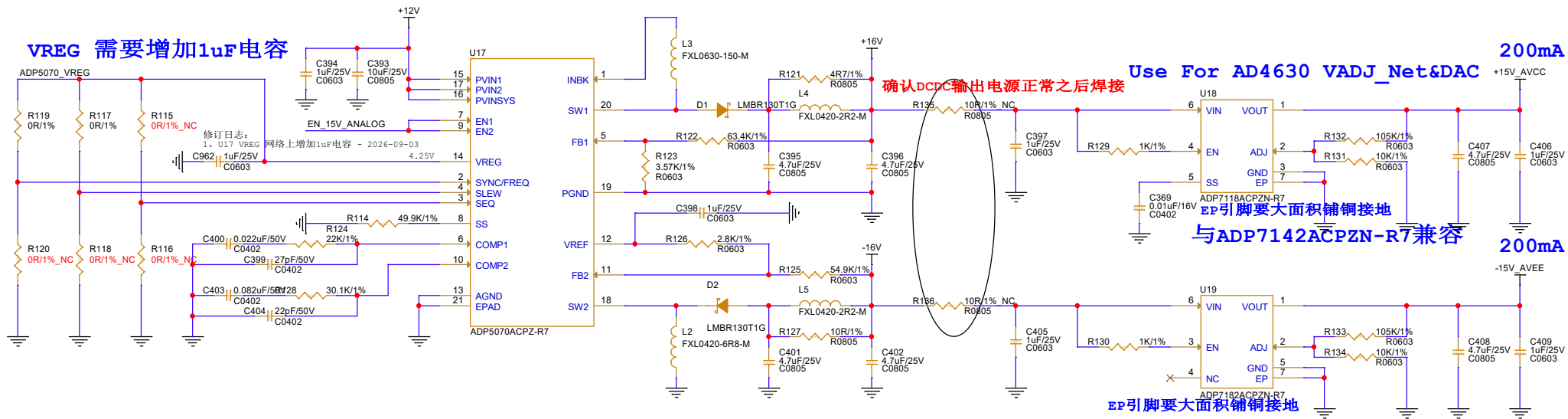


VADJ_VCCO

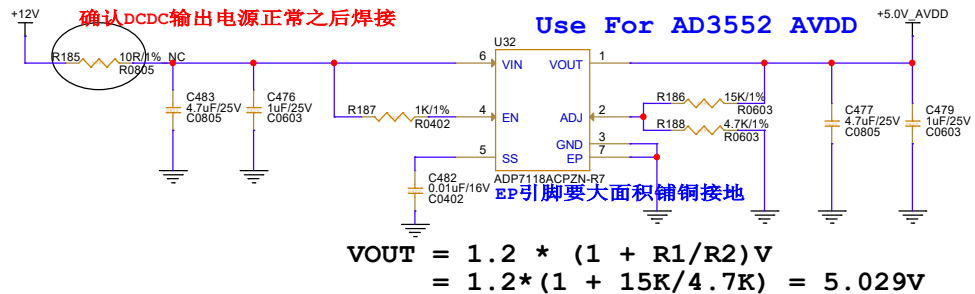
EN_3V3_VCCO	<<EN_3V3_VCCO	[27]
EN_1V8_VCCO	<<EN_1V8_VCCO	[27]
EN_2V5_VCCO	<<EN_2V5_VCCO	[27]
EN_1V2_VCCO	<<EN_1V2_VCCO	[27]
EN_2V5_VPP	<<EN_2V5_VPP	[27]

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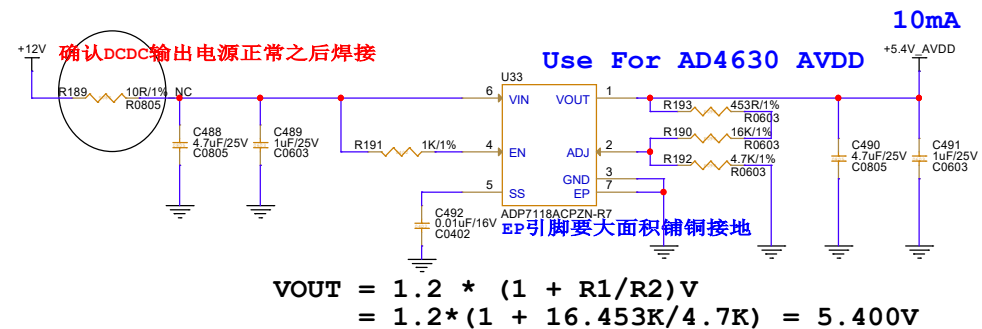
PMIC ANALOG



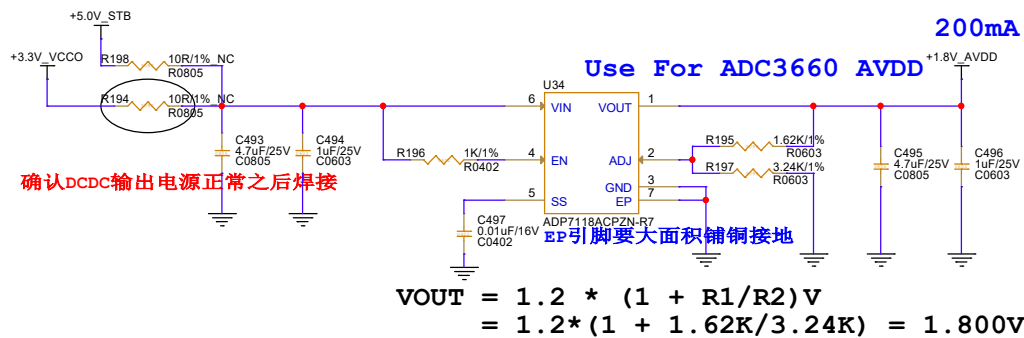
$$\text{SSTIME } (\mu\text{s}) = 380 \mu\text{s} + 0.6 \times \text{CSS}$$



$$\begin{aligned} V_{OUT} &= 1.2 * (1 + R1/R2) V \\ &= 1.2 * (1 + 15K/4.7K) = 5.029V \end{aligned}$$



$$\begin{aligned} V_{OUT} &= 1.2 * (1 + R_1/R_2) V \\ &= 1.2 * (1 + 16.453K/4.7K) = 5.400V \end{aligned}$$

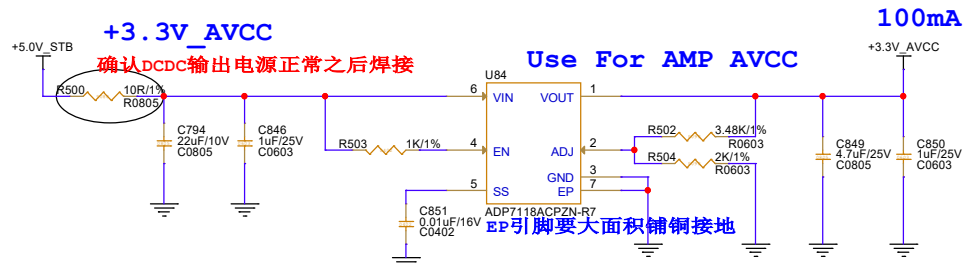


$$\begin{aligned} V_{OUT} &= 1.2 * (1 + R1/R2) V \\ &= 1.2 * (1 + 1.62K/3.24K) = 1.800V \end{aligned}$$

EN_15V_ANALOG << EN_15V_ANALOG [26]

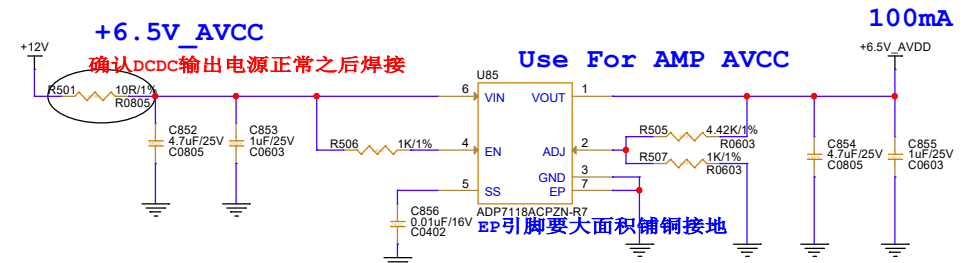
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PMIC_AMP



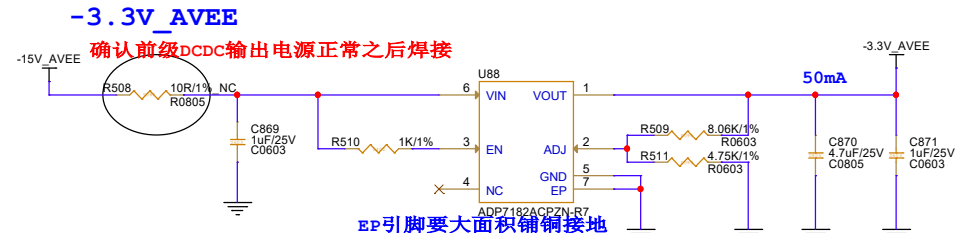
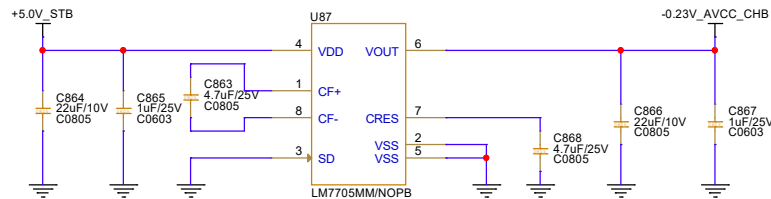
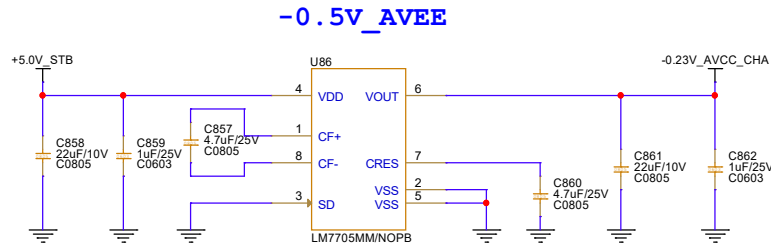
$$VOUT = 1.2 * (1 + R1/R2)V$$

$$= 1.2 * (1 + 3.48K/2K) = 3.288V$$



$$VOUT = 1.2 * (1 + R1/R2)V$$

$$= 1.2 * (1 + 4.42K/1K) = 6.504V$$

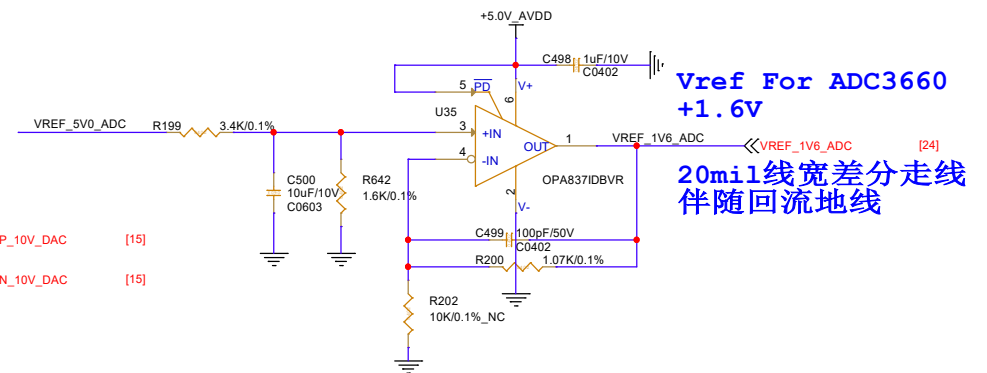
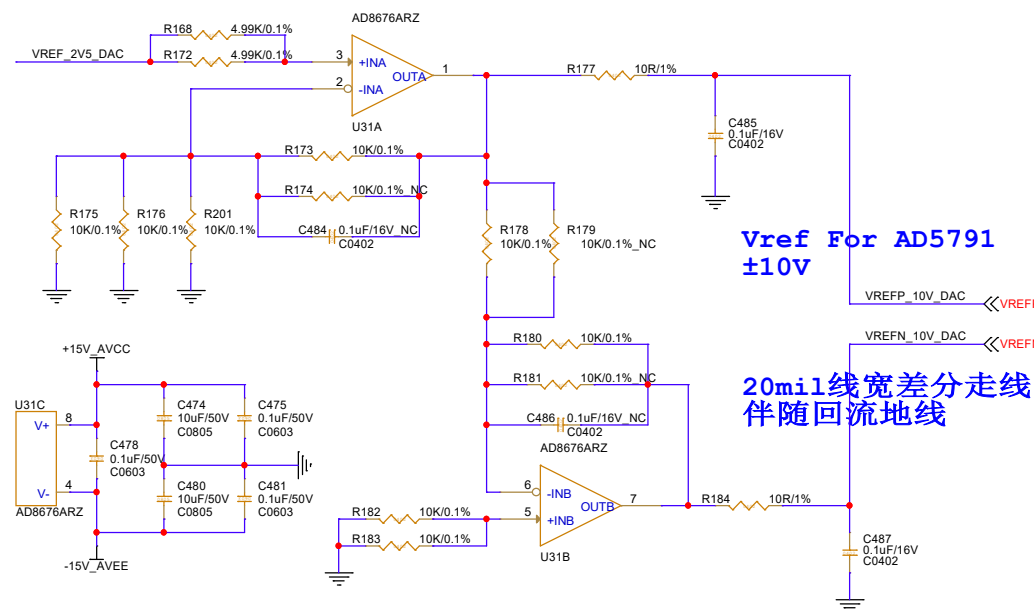
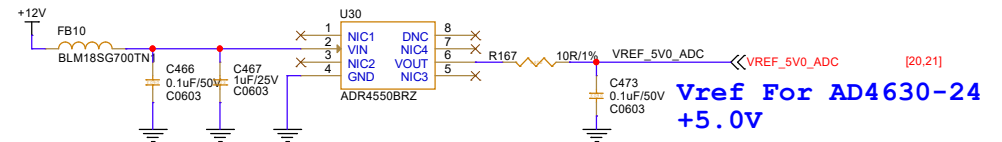
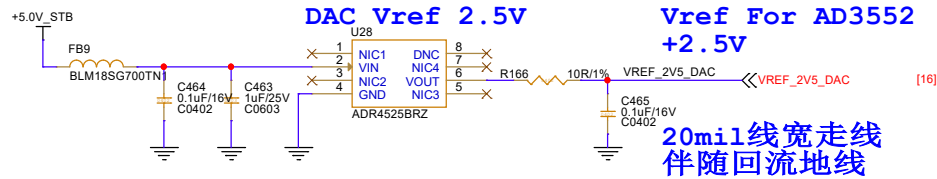


$$-VOUT = -1.22 * (1 + RFB1/RFB2)$$

$$= -1.22 * (1 + 8.06K/4.75K) = -3.29V$$

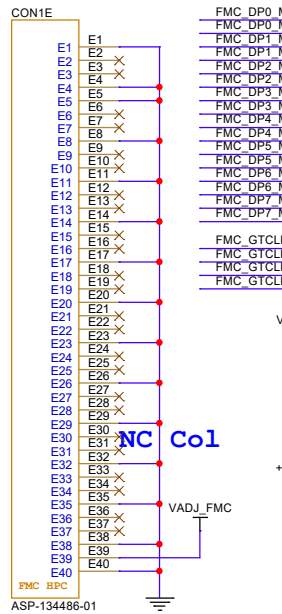
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PMIC_VREF

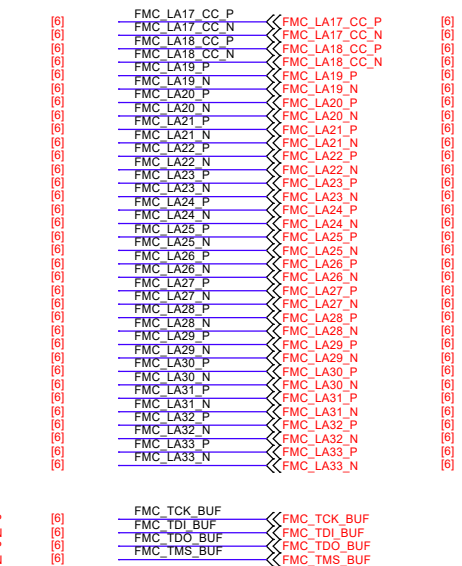
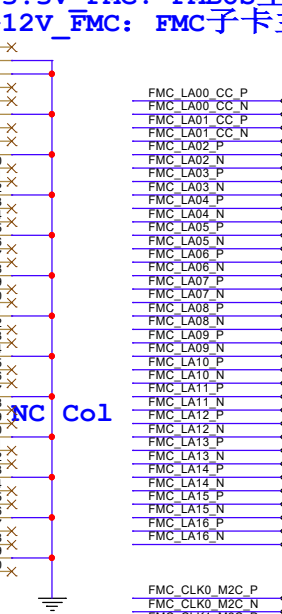


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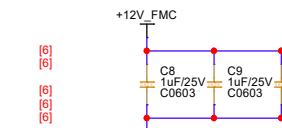
参考ZCU102 HPC0



VADJ FMC: IO/Logic 电源
+3.3V FMC: PMBUS主电源 (常电)
+12V FMC: FMC子卡主电源



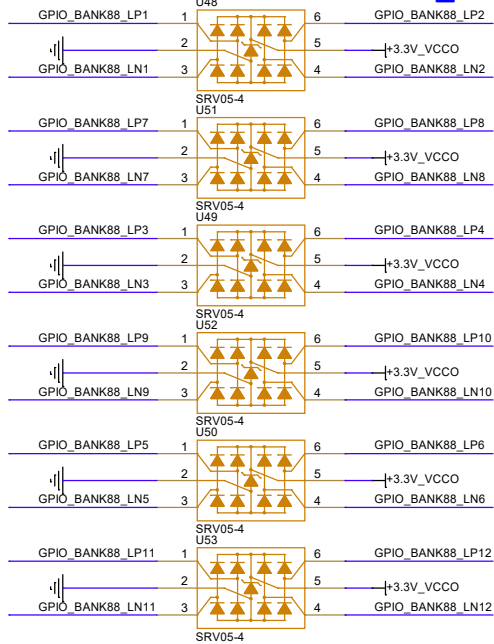
DIFF/Single IO
差分阻抗: 100R
单端阻抗: 50R



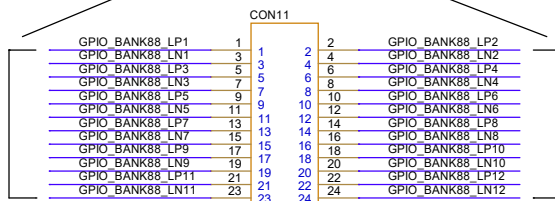
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CON_HEADER&RS485&RS422&RS232

ESD Array For Bank88 +3.3V_VCCO



丝印标注:
HDIO Bank8
3.3V IO



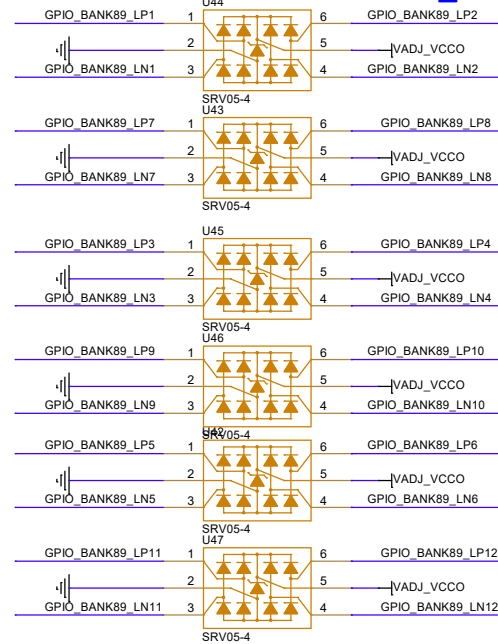
HEADER_2X12P_2R54



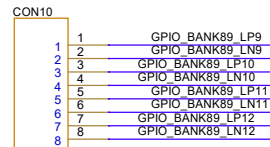
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丝印标注:
HDIO Bank89
2.5V/3.3V IO

ESD Array For Bank89 VADJ_VCCO



丝印标注:
HDIO Bank89
2.5V/3.3V IO



HEADER_1X8P_2R54

BANK88 3.3V Level

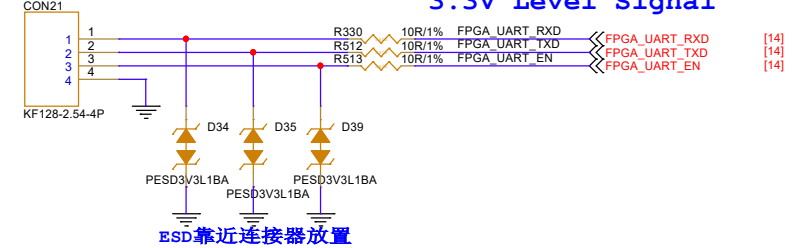


BANK89 2.5V or 3.3V Level

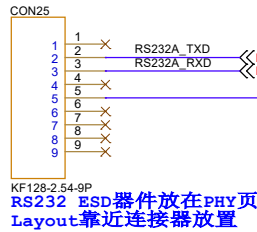


Bank88 (3.3V) 引出来24个引脚, Bank89引出来16个引脚,
一共是40个引脚, 用20*2p的排针引出;
将bank89剩下的8个引脚用一个8*1p的排针引出来。

3.3V Level Signal

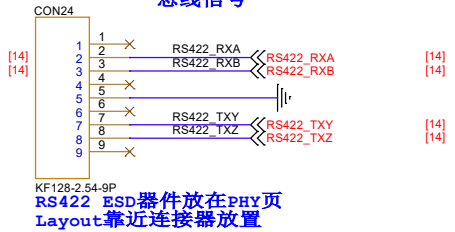


ESD靠近连接器放置

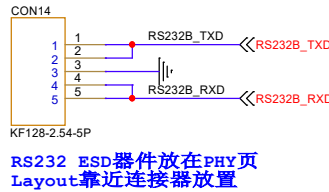


RS232 ESD器件放在PHY页
Layout靠近连接器放置

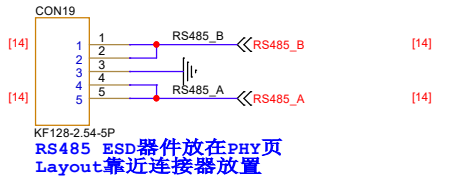
总线信号



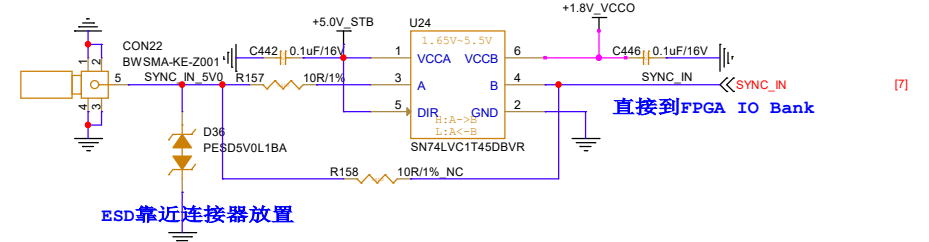
RS422 ESD器件放在PHY页
Layout靠近连接器放置



RS232 ESD器件放在PHY页
Layout靠近连接器放置



RS485 ESD器件放在PHY页
Layout靠近连接器放置

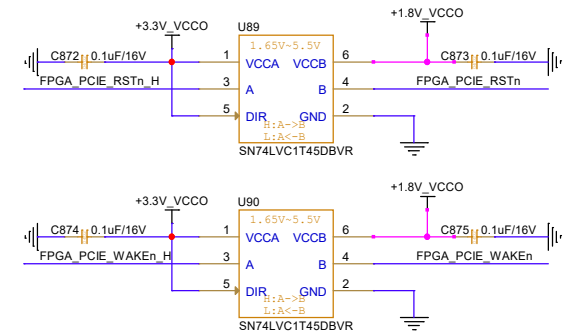
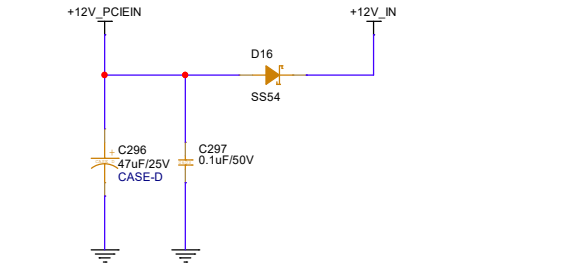
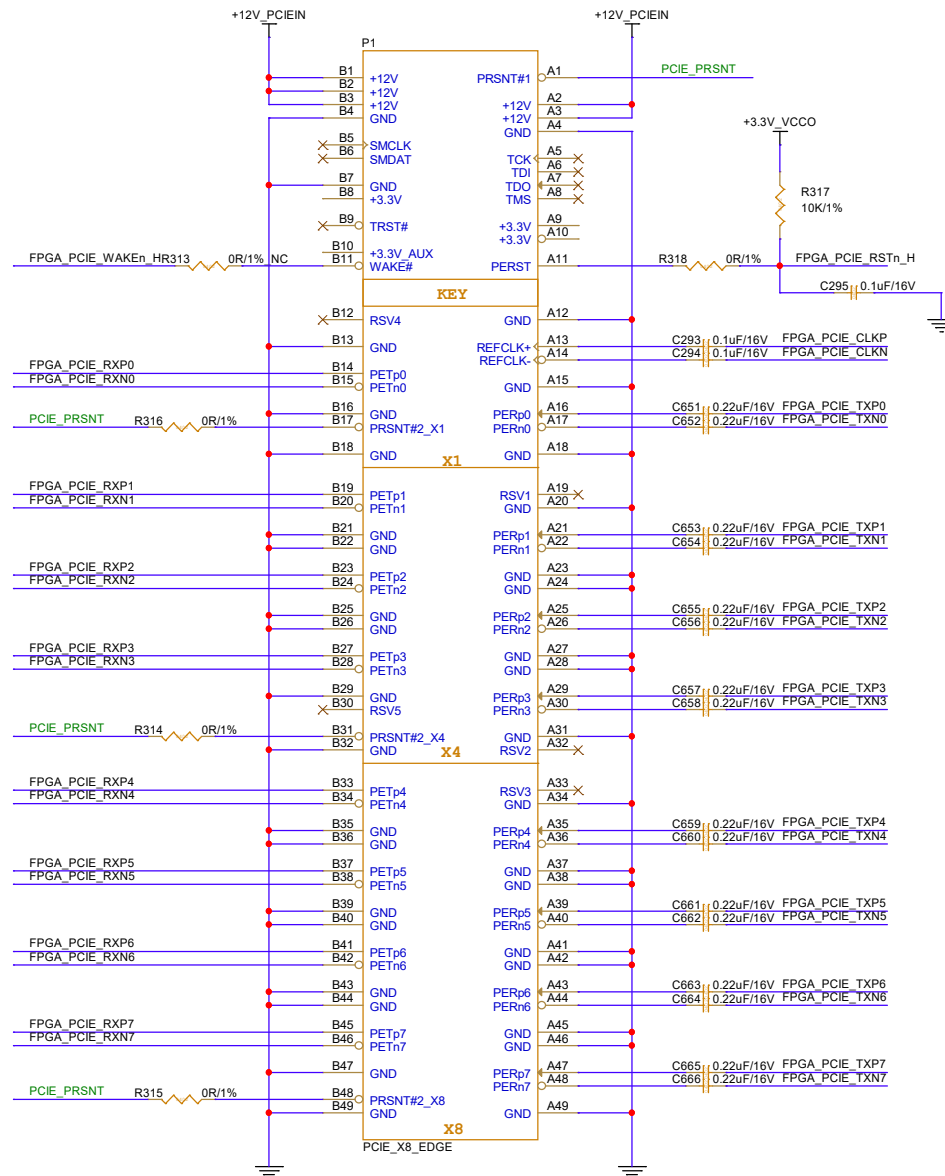


ESD靠近连接器放置

直接到FPGA IO Bank

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CON_PCIE_GOLDEN_FINGER



FPGA_PCIE_WAKEn	FPGA_PCIE_WAKEn	[7]
FPGA_PCIE_RSTn	FPGA_PCIE_RSTn	[7]
FPGA_PCIE_CLKP	FPGA_PCIE_CLKP	[8]
FPGA_PCIE_CLKN	FPGA_PCIE_CLKN	[8]
FPGA_PCIE_TXP[0:7]	FPGA_PCIE_TXP[0:7]	[8]
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FPGA_PCIE_RXP[0:7]	FPGA_PCIE_RXP[0:7]	[8]
FPGA_PCIE_RXN[0:7]	FPGA_PCIE_RXN[0:7]	[8]

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